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1 Misc

1.1 Default Code [24a798]

```
1 #include <bits/stdc++.h>
2 using namespace std;
3 #define int long long
4 #define debug(a...) cerr << #a << " = ", dout(a)
5 void dout() { cerr << "\n"; }
6 template <typename A, typename... B>
7 void dout(A a, B... b) { cerr << a << ' ', dout(b...); }
8
9 void solve(){
10
11 }
12
13 signed main(){
14     ios::sync_with_stdio(0), cin.tie(0);
15
16     int t = 1;
17     while (t--){
18         solve();
19     }
20
21     return 0;
22 }
```

1.2 Run.py

```
1 from os import *
2 f = "pA"
3 while 1:
4     i = input("> ") or f; system("clear"); f = i
5     print(f"file = {f}")
6     if system(f"g++ {f}.cpp -std=c++17 -Wall -Wextra -Wshadow
7         -O2 -D LOCAL -g -fsanitize=undefined,address -o {f}
8         "):
9         print("CE"); continue
10    for x in sorted(listdir()):
11        if x.startswith(f) and x.endswith(".in"):
12            print(x); system(f"./{f} < {x}") and print("RE");
13            print()
```

1.3 Diff

```
1 set -e
2 g++ ac.cpp -o ac
3 g++ wa.cpp -o wa
4 for ((i=0;i++))
5 do
6     echo "$i"
7     python3 gen.py > input
8     ./ac < input > ac.out
9     ./wa < input > wa.out
10    diff ac.out wa.out || break
11 done
```

1.4 Hash Command

```
1 cat file.cpp | cpp -dD -P -fpreprocessed | tr -d "[:space:]"
2 | md5sum | cut -c-6
```

1.5 Run.sh

```
1 if [ ! -f $1 ] || [ $1.cpp -nt $1 ]; then
2     g++ $1.cpp -std=c++17 -Wall -Wextra -Wshadow -O2 -D LOCAL
3     -g -fsanitize=undefined,address -o $1 ||
4     { echo "Compile failed."; exit 1; }
5 fi
6
7 ./ $1
```

1.6 Run.bat

```
1 @echo off
2 for /f %i in ('dir /b /od %1.cpp %1.exe 2^>nul') do set L=%i
3 if not "%L%"=="%1.exe" (
4     g++ %1.cpp -std=c++17 -Wall -Wextra -Wshadow -O2 -D LOCAL ^
5     -g -o %1 || (echo Compile failed.& exit /b 1)
6 )
7 %1
```

1.7 Custom Set PQ Sort [d4df55]

```
1 // 所有自訂的結構體，務必檢查相等的 case，給所有元素一個排序
2 // 的依據
3 struct my_struct{
4     int val;
5     my_struct(int _val) : val(_val) {}
6 };
7
8 auto cmp = [](my_struct a, my_struct b) {
9     return a.val > b.val;
10 };
11
12 set<my_struct, decltype(cmp)> ss({1, 2, 3}, cmp);
13 priority_queue<my_struct, vector<my_struct>, decltype(cmp)>
14     pq(cmp, {1, 2, 3});
15 map<my_struct, my_struct, decltype(cmp)> mp({{1, 4}, {2, 5},
16     {3, 6}}, cmp);
```

1.8 Dynamic Bitset [c78aa8]

```
1 const int MAXN = 2e5 + 5;
2 template <int len = 1>
3 void solve(int n) {
4     if (n > len) {
5         solve<min(len*2, MAXN)>(n);
6         return;
7     }
8     bitset<len> a;
9 }
```

1.9 Enumerate Subset [a13e46]

```
1 // 時間複雜度 O(3^n)
2 // 枚舉每個 mask 的子集
3 for (int mask=0; mask<(1<<n); mask++){
4     for (int s=mask; s>=0; s=(s-1)&m){
5         // s 是 mask 的子集
6         if (s==0) break;
7     }
8 }
```

1.10 Fast Input [6f8879]

```
1 // fast IO
2 // 6f8879
3 inline char readchar(){
4     static char buffer[BUFSIZ], *now = buffer + BUFSIZ, *
5     end = buffer + BUFSIZ;
6     if (now == end)
7     {
8         if (end < buffer + BUFSIZ)
9             return EOF;
10        end = (buffer + fread(buffer, 1, BUFSIZ, stdin));
11        now = buffer;
12    }
13    return *now++;
14 }
15 inline int nextint(){
16     int x = 0, c = readchar(), neg = false;
17     while(('0' > c || c > '9') && c!='-' && c!=EOF) c =
18         readchar();
19     if(c == '-') neg = true, c = readchar();
20     while('0' <= c && c <= '9') x = (x<<3) + (x<<1) + (c^'0')
21         , c = readchar();
22     if(neg) x = -x;
23     return x; // returns 0 if EOF
24 }
```

1.11 Grid paper

```
1 i+<esc>25A---+<esc>
2 o|<esc>25A |<esc>
3 ggVGyG35pGdd
```

1.12 OEIS [25cf74]

```
1 // 若一個線性遞迴有 k 項，給他恰好 2*k 個項可以求出線性遞迴
2 // s = 1 1 3 7 17，則它會回傳 1 2
3 vector<int> BerlekampMassey(vector<int> s) {
4     if (s.empty()) return vector<int>();
5     int n = s.size(), L = 0, m = 0;
6     vector<int> C(n), B(n), T;
7     C[0] = B[0] = 1;
8
9     int b = 1;
10    for (int i=0; i<n; i++) {
11        ++m; int d = s[i]%MOD;
12        for (int j=1; j<L+1; j++) d = (d+C[j]*s[i-j])%MOD;
13        if (!d) continue;
14        T = C; int coef = d*qp(b, MOD-2, MOD)%MOD;
15        for (int j=m; j<n; j++) C[j] = (C[j]-coef*B[j-m])%
16            MOD;
17        if (2*L>i) continue;
18        L = i+1-L; B=T; b=d; m=0;
19    }
20    C.resize(L+1); C.erase(C.begin());
21    for (int &x : C) x = (MOD-x)%MOD;
22    reverse(C.begin(), C.end());
23    return C;
24 }
```

1.13 OEIS

```
1 from fractions import Fraction as F
2 def BM(s):
```

```

3   n = len(s); L=m=0; C=[F(0)]*n; B=[F(0)]*n; C[0]=B[0]=F(1)
4       ; b=F(1)
5   for i in range(n):
6       m+=1; d=sum(C[j]*s[i-j] for j in range(L+1))
7       if d==0: continue
8       T = C[:]; coef = F(d/b)
9       for j in range(m, n): C[j] -= coef*B[j-m]
10      if 2*L<=i: L, B, b, m = i+1-L, T, d, 0
11      return [-x for x in C[1:L+1]]

```

1.14 Pragma [09d13e]

```

1 #pragma GCC optimize("O3,unroll-loops")
2 #pragma GCC target("avx,avx2,sse,sse2,sse3,sse4,popcnt")

```

1.15 Python

```

1 # Decimal
2 from decimal import *
3 getcontext().prec = 6
4
5 # system setting
6 sys.setrecursionlimit(100000)
7 sys.set_int_max_str_digits(10000)
8
9 # turtle
10 from turtle import *
11
12 N = 3000000010
13 setworldcoordinates(-N, -N, N, N)
14 hideturtle()
15 speed(100)
16
17 def draw_line(a, b, c, d):
18     teleport(a, b)
19     goto(c, d)
20
21 def write_dot(x, y, text, diff=1): # diff = 文字的偏移
22     teleport(x, y)
23     dot(5, "red")
24
25     teleport(x+N/100*diff, y+N/100*diff)
26     write(text, font=("Arial", 5, "bold"))
27
28 # usage
29 draw_line(*a[i], *(a[i-1]))
30 write_dot(*a[i], str(a[i]))
31 # 以自己左方 100 單位為圓心向前畫一個 90 度的弧 / 正五邊形
   (參數可以是負的)
32 circle(100, extent = 90)
33 circle(100, steps = 5)
34 left(90) # 左轉 90 度
35 fd(-100) # 倒退 100 單位
36 penup()
37 pendown()
38
39 tracer(0, 0) # IO 優化 · 必須在程式碼最後手動 update()
40 pos() # 回傳目前座標
41 heading() # 回傳目前方向: 0 是右方、90 是上方
42 color("red") # 改變顏色 · 可以用文字或是 #000000 的格式
43
44 # OOP

```

```

45 class Point:
46     def __init__(self, x, y):
47         self.x = x
48         self.y = y
49
50     def __add__(self, o): # use dir(int) to know operator
51         name
52         return Point(self.x+o.x, self.y+o.y)
53
54     @property
55     def distance(self):
56         return (self.x**2 + self.y**2)**(0.5)
57
58 a = Point(3, 4)
59 print(a.distance)
60
61 # Fraction
62 from fractions import Fraction
63 a = Fraction(Decimal(1.1))
64 a.numerator # 分子
65 a.denominator # 分母

```

1.16 Xor Basis [123f2d]

```

1 vector<int> basis;
2 void add_vector(int x){
3     for (auto v : basis){
4         x=min(x, x^v);
5     }
6     if (x) basis.push_back(x);
7 }
8
9 // 給一數字集合 S · 求能不能 XOR 出 x
10 bool check(int x){
11     for (auto v : basis){
12         x=min(x, x^v);
13     }
14     return (x == 0);
15 }
16
17 // 給一數字集合 S · 求能 XOR 出多少數字
18 // 答案等於 2^{basis 的大小}
19
20 // 給一數字集合 S · 求 XOR 出最大的數字
21 int get_max(){
22     int ans=0;
23     for (auto v : basis){
24         ans=max(ans, ans^v);
25     }
26     return ans;
27 }

```

1.17 disable ASLR [29a49f]

```

1 // Disable randomization of memory addresses
2 // setarch $(uname -m) -R /bin/bash
3
4 #include <sys/personality.h>
5 #include <bits/stdc++.h>
6 int main(int argc, char* argv[]){
7     personality(ADDR_NO_RANDOMIZE);
8     execv(argv[1], argv+1);
9 }

```

1.18 hash windows

```

1 def get_hash(path):
2     from subprocess import run, PIPE
3     from hashlib import md5
4
5     p = run(
6         ["cpp", "-dD", "-P", "-fpreprocessed", path],
7         stdout = PIPE,
8         stderr = PIPE,
9         text = True
10    )
11
12    if p.returncode != 0:
13        raise RuntimeError(p.stderr)
14
15    s = ''.join(p.stdout.split())
16    ret = md5(s.encode()).hexdigest()
17    return ret[:6]
18
19 print(get_hash("Suffix_Array.cpp"))

```

1.19 random int [9cc603]

```

1 mt19937 seed(chrono::steady_clock::now().time_since_epoch().
2     count());
3 int rng(int l, int r){
4     return uniform_int_distribution<int>(l, r)(seed);
5 }

```

2 Convolution

2.1 FFT [16591a]

```

1 typedef complex<double> cd;
2
3 // 778272
4 void FFT(vector<cd> &a) {
5     int n = a.size(), L = 31-__builtin_clz(n);
6     vector<complex<long double>> R(2, 1);
7     vector<cd> rt(2, 1);
8     for (int k=2 ; k<n ; k*=2){
9         R.resize(n), rt.resize(n);
10        auto x = polar(1.0L, acos(-1.0L) / k);
11        for (int i=k ; i<2*k ; i++) rt[i] = R[i] = (i&1 ? R[i/2]*x : R[i/2]);
12    }
13
14    vector<int> rev(n);
15    for (int i=0 ; i<n ; i++) rev[i] = (rev[i/2] | (i&1)<<L) /2;
16    for (int i=0 ; i<n ; i++) if (i<rev[i]) swap(a[i], a[rev[i]]);
17    for (int k=1 ; k<n ; k*=2){
18        for (int i=0 ; i<n ; i+=2*k){
19            for (int j=0 ; j<k ; j++){
20                // cd z = rt[j+k] * a[i+j+k];
21                auto x = (double *)&rt[j+k], y = (double *)&a[i+j+k];
22                cd z(x[0]*y[0] - x[1]*y[1], x[0]*y[1] + x[1]*y[0]);
23                a[i+j+k] = a[i+j]-z;
24                a[i+j] += z;
25            }
26        }
27    }
28 }

```

```

27 }
28 }
29
30 // 192528
31 vector<double> PolyMul(const vector<double> a, const vector<
double> b){
32     if (a.empty() || b.empty()) return {};
33     vector<double> res(a.size()+b.size()-1);
34     int L = 32 - __builtin_clz(res.size()), n = 1<<L;
35     vector<cd> in(n), out(n);
36     copy(a.begin(), a.end(), in.begin());
37     for (int i=0 ; i<b.size() ; i++) in[i].imag(b[i]);
38     FFT(in);
39     for (cd& x : in) x *= x;
40     for (int i=0 ; i<n ; i++) out[i] = in[-i & (n - 1)] -
conj(in[i]);
41     FFT(out);
42     for (int i=0 ; i<res.size() ; i++) res[i] = imag(out[i])
/ (4 * n);
43     return res;
44 }

```

2.2 FFT any mod [412000]

```

1 // n=524288, ~400ms
2 const int MOD = 998244353;
3 typedef complex<double> cd;
4
5 // bb6d94
6 void FFT(vector<cd> &a) {
7     int n = a.size(), L = 31-__builtin_clz(n);
8     vector<complex<long double>> R(2, 1);
9     vector<cd> rt(2, 1);
10    for (int k=2 ; k<n ; k*=2){
11        R.resize(n);
12        rt.resize(n);
13        auto x = polar(1.0/L, acos(-1.0/L) / k);
14        for (int i=k ; i<2*k ; i++){
15            rt[i] = R[i] = (i&1 ? R[i/2]*x : R[i/2]);
16        }
17    }
18
19    vector<int> rev(n);
20    for (int i=0 ; i<n ; i++) rev[i] = (rev[i/2] | (i&1)<<L)
/2;
21    for (int i=0 ; i<n ; i++) if (i<rev[i]) swap(a[i], a[rev[
i]]);
22    for (int k=1 ; k<n ; k*=2){
23        for (int i=0 ; i<n ; i+=2*k){
24            for (int j=0 ; j<k ; j++){
25                auto x = (double *)&rt[j+k];
26                auto y = (double *)&a[i+j+k];
27                cd z(x[0]*y[0] - x[1]*y[1], x[0]*y[1] + x[1]*
y[0]);
28                a[i+j+k] = a[i+j]-z;
29                a[i+j] += z;
30            }
31        }
32    }
33 }
34
35 // c3dcf6
36 vector<int> PolyMul(vector<int> a, vector<int> b){
37     if (a.empty() || b.empty()) return {};
38     vector<int> res(a.size()+b.size()-1);

```

```

39 int B = 32-__builtin_clz(res.size()), n = (1<<B), cut = (
int)(sqrt(MOD));
40 vector<cd> L(n), R(n), outs(n), outl(n);
41
42 for (int i=0 ; i<a.size() ; i++) L[i] = cd((int) a[i]/cut
, (int)a[i]%cut);
43 for (int i=0 ; i<b.size() ; i++) R[i] = cd((int) b[i]/cut
, (int)b[i]%cut);
44 FFT(L), FFT(R);
45 for (int i=0 ; i<n ; i++){
46     int j = -i&(n-1);
47     outl[j] = (L[i]+conj(L[j])) * R[i]/(2.0*n);
48     outs[j] = (L[i]-conj(L[j])) * R[i]/(2.0*n)/1i;
49 }
50 FFT(outl), FFT(outs);
51 for (int i=0 ; i<res.size() ; i++){
52     int av = (int)(real(outl[i]+0.5), cv = (int)(imag(
outs[i]+0.5);
53     int bv = (int)(imag(outl[i]+0.5) + (int)(real(outs[i
]+0.5);
54     res[i] = ((av%MOD*cut+bv) % MOD*cut+cv) % MOD;
55 }
56 return res;
57 }

```

2.3 FWT [e50788]

```

1 // 已經把 mint 刪掉 · 需要增加註解
2 vector<int> xor_convolution(vector<int> a, vector<int> b, int
k) {
3     if (k==0) return vector<int>{a[0]*b[0]};
4     vector<int> aa(1 << (k - 1)), bb(1 << (k - 1));
5     for (int i=0 ; i<(1<<(k-1)) ; i++){
6         aa[i] = a[i] + a[i + (1 << (k - 1))];
7         bb[i] = b[i] + b[i + (1 << (k - 1))];
8     }
9     vector<int> X = xor_convolution(aa, bb, k - 1);
10    for (int i=0 ; i<(1<<(k-1)) ; i++){
11        aa[i] = a[i] - a[i + (1 << (k - 1))];
12        bb[i] = b[i] - b[i + (1 << (k - 1))];
13    }
14    vector<int> Y = xor_convolution(aa, bb, k - 1);
15    vector<int> c(1 << k);
16    for (int i=0 ; i<(1<<(k-1)) ; i++){
17        c[i] = (X[i] + Y[i]) / 2;
18        c[i + (1 << (k - 1))] = (X[i] - Y[i]) / 2;
19    }
20    return c;
21 }

```

2.4 Min Convolution Concave Concave [ffb28d]

```

1 // 需要增加註解
2 // min convolution
3 vector<int> mkk(vector<int> a, vector<int> b) {
4     vector<int> slope;
5     FOR (i, 1, ssize(a) - 1) slope.pb(a[i] - a[i - 1]);
6     FOR (i, 1, ssize(b) - 1) slope.pb(b[i] - b[i - 1]);
7     sort(all(slope));
8     slope.insert(begin(slope), a[0] + b[0]);
9     partial_sum(all(slope), begin(slope));
10    return slope;
11 }

```

2.5 NTT mod 998244353 [976c8d]

```

1 const int MOD = (119 << 23) + 1, ROOT = 62; // = 998244353
2 // For p < 2^30 there is also e.g. 5 << 25, 7 << 26, 479 <<
21
3 // and 483 << 21 (same root). The last two are > 10^9.
4
5 // 7e041b
6 vector<int> rt(2, 1);
7 void NTT_init(int n){
8     static bool lastInit = 0;
9     if (lastInit<n){
10         lastInit = n;
11         for (int k=2, s=2 ; k<n ; k*=2, s++){
12             rt.resize(n);
13             int z[] = {1, qp(ROOT, MOD>>s, MOD)};
14             for (int i=k ; i<2*k ; i++) rt[i] = rt[i/2]*z[i
&1]%MOD;
15         }
16     }
17 }
18
19 // fec4b8
20 void NTT(vector<int> &a, int isInverse = false) {
21     int n = a.size();
22     int L = 31-__builtin_clz(n);
23     NTT_init(n);
24
25     vector<int> rev(n);
26     for (int i=0 ; i<n ; i++) rev[i] = (rev[i/2]|(i&1)<<L)/2;
27     for (int i=0 ; i<n ; i++){
28         if (i<rev[i]) swap(a[i], a[rev[i]]);
29     }
30
31     for (int k=1 ; k<n ; k*=2){
32         for (int i=0 ; i<n ; i+=2*k){
33             for (int j=0 ; j<k ; j++){
34                 int z = rt[j+k]*a[i+j+k]%MOD, &ai = a[i+j];
35                 a[i+j+k] = ai-z+(z>ai ? MOD : 0);
36                 ai += (ai+z>MOD ? z-MOD : z);
37             }
38         }
39     }
40
41     if (isInverse) {
42         reverse(a.begin()+1, a.end());
43         int n_inv = qp(n, MOD-2, MOD);
44         for (int i=0 ; i<n ; i++) a[i] = a[i]*n_inv%MOD;
45     }
46 }
47
48 // 771576
49 vector<int> polyMul(vector<int> a, vector<int> b){
50     if (a.empty() || b.empty()) return {};
51     int s = a.size()+b.size()-1, B = 32-__builtin_clz(s), n =
1<<B;
52     vector<int> L(a), R(b), out(n);
53     L.resize(n), R.resize(n);
54     NTT(L), NTT(R);
55     for (int i=0 ; i<n ; i++) out[i] = L[i]*R[i]%MOD;
56     NTT(out, true);
57     out.resize(s);
58     return out;
59 }

```

2.6 PolyInv [b388fb]

```

1 vector<int> polyInv(vector<int> a){
2     int n = a.size();
3     vector<int> ret(1, qp(a[0], MOD-2, MOD));
4
5     for (int m=1; m<n; m<=1){
6         if (n<2*m) a.resize(2*m);
7         vector<int> v1(a.begin(), a.begin()+2*m), v2 = ret;
8         v1.resize(4*m), v2.resize(4*m);
9         NTT(v1), NTT(v2);
10        for (int i=0; i<4*m; i++) v1[i] = v1[i]*v2[i]%MOD*
            v2[i]%MOD;
11        NTT(v1, true);
12        ret.resize(2*m);
13        for (int i=0; i<m; i++) ret[i] = (ret[i]+ret[i])%
            MOD;
14        for (int i=0; i<2*m; i++) ret[i] = ((ret[i]-v1[i])%
            MOD+MOD)%MOD;
15    }
16    ret.resize(n);
17    return ret;
18 }

```

2.7 PolySqrt [1ea321]

```

1 // ---- sqrt for formal power series over MOD=998244353 ----
2 int mod_sqrt(int a) { // Tonelli-Shanks; return -1 if no
    solution
3     a %= MOD; if (a < 0) a += MOD;
4     if (a == 0) return 0;
5     if (qp(a, (MOD - 1) / 2, MOD) == MOD - 1) return -1; //
        non-residue
6     if (MOD % 4 == 3) return qp(a, (MOD + 1) / 4, MOD);
7     int q = MOD - 1, s = 0;
8     while ((q & 1) == 0) q >>= 1, ++s;
9     int z = 2;
10    while (qp(z, (MOD - 1) / 2, MOD) != MOD - 1) ++z; // find
        non-residue
11    long long c = qp(z, q, MOD);
12    long long x = qp(a, (q + 1) / 2, MOD);
13    long long t = qp(a, q, MOD);
14    int m = s;
15    while (t != 1) {
16        int i = 1;
17        long long tt = t * t % MOD;
18        while (tt != 1) {
19            tt = tt * tt % MOD;
20            ++i;
21            if (i == m) return -1; // shouldn't happen
22        }
23        long long b = qp(c, 1LL << (m - i - 1), MOD);
24        x = x * b % MOD;
25        long long b2 = b * b % MOD;
26        t = t * b2 % MOD;
27        c = b2;
28        m = i;
29    }
30    return (int)x;
31 }
32
33 vector<int> polySqrt(vector<int> a) {
34     int n = (int)a.size();
35
36     // all zero -> zero

```

```

37 bool all0 = true;
38 for (int x : a) if (x % MOD != 0) { all0 = false; break;
39 }
40 if (all0) return vector<int>(n, 0);
41
42 // first nonzero position
43 int k = 0;
44 while (k < n && a[k] == 0) ++k;
45 if (k & 1) return {}; // odd -> impossible
46 int shift = k / 2;
47
48 // strip x^k
49 vector<int> f;
50 f.reserve(n - k);
51 for (int i = k; i < n; ++i) f.push_back(a[i]);
52
53 // constant term sqrt, choose canonical sign
54 int c0 = (f.empty() ? 0 : f[0] % MOD);
55 int s0 = mod_sqrt(c0);
56 if (s0 == -1) return {};
57 if (s0 > MOD - s0) s0 = MOD - s0; // choose the smaller
        nonnegative root
58
59 // We must produce N terms after placing the shift =>
        need = N - shift
60 int need = n - shift;
61 if ((int)f.size() < need) f.resize(need, 0);
62
63 // Newton: g_{new} = (g + f/g) / 2 (mod x^m)
64 const int INV2 = (MOD + 1) / 2;
65 vector<int> g(1, s0);
66
67 for (int m = 1; m < need; m <= 1) {
68     int lim = min(need, m << 1);
69
70     vector<int> g_cut = g; g_cut.resize(lim);
71     vector<int> inv_g = polyInv(g_cut); // inv up
        to lim
72     vector<int> f_cut(f.begin(), f.begin() + lim);
73
74     vector<int> q = polyMul(inv_g, f_cut);
75     if ((int)q.size() > lim) q.resize(lim);
76
77     g.resize(lim);
78     for (int i = 0; i < lim; ++i) {
79         long long gi = (i < (int)g_cut.size() ? g_cut[i]
80             : 0);
81         long long qi = (i < (int)q.size() ? q[i] : 0);
82         g[i] = (gi + qi) % MOD * 1LL * INV2 % MOD;
83     }
84     g.resize(need);
85
86     // place back the x^{k/2} shift
87     vector<int> res(n, 0);
88     for (int i = 0; i < need && i + shift < n; ++i) res[i +
        shift] = g[i];
89     return res;

```

3 Data-Structure

3.1 BIT [7ef3a9]

```

1 vector<int> BIT(MAX_SIZE);
2

```

```

3 // const int MAX_N = (1<<20)
4 int k_th(int k) { // 回傳 BIT 中第 k 小的元素 (based-1)
5     int res = 0;
6     for (int i=MAX_N>>1; i>=1; i>=1)
7         if (BIT[res+i]<k)
8             k -= BIT[res+i];
9     return res+1;
10 }

```

3.2 Disjoint Set Persistent [447002]

```

1 struct Persistent_Disjoint_Set{
2     Persistent_Segment_Tree arr, sz;
3
4     void init(int n){
5         arr.init(n);
6         vector<int> v1;
7         for (int i=0; i<n; i++){
8             v1.push_back(i);
9         }
10        arr.build(v1, 0);
11
12        sz.init(n);
13        vector<int> v2;
14        for (int i=0; i<n; i++){
15            v2.push_back(1);
16        }
17        sz.build(v2, 0);
18    }
19
20    int find(int a){
21        int res = arr.query_version(a, a+1, arr.version.size()
            -1).val;
22        if (res==a) return a;
23        return find(res);
24    }
25
26    bool unite(int a, int b){
27        a = find(a);
28        b = find(b);
29
30        if (a!=b){
31
32            int sz1 = sz.query_version(a, a+1, arr.version.
            size()-1).val;
33            int sz2 = sz.query_version(b, b+1, arr.version.
            size()-1).val;
34
35            if (sz1<sz2){
36                arr.update_version(a, b, arr.version.size()
                    -1);
37                sz.update_version(b, sz1+sz2, arr.version.
                    size()-1);
38            }else{
39                arr.update_version(b, a, arr.version.size()
                    -1);
40                sz.update_version(a, sz1+sz2, arr.version.
                    size()-1);
41            }
42            return true;
43        }
44        return false;
45    }
46 };

```

3.3 PBDS GP Hash Table [866cf6]

```
1 #include <ext/pb_ds/assoc_container.hpp>
2 using namespace __gnu_pbds;
3 typedef tree<int, null_type, less<int>, rb_tree_tag,
4   tree_order_statistics_node_update> order_set;
5 struct custom_hash {
6   static uint64_t splitmix64(uint64_t x) {
7     // http://xorshift.di.unimi.it/splitmix64.c
8     x += 0x9e3779b97f4a7c15;
9     x = (x ^ (x >> 30)) * 0xbf58476d1ce4e5b9;
10    x = (x ^ (x >> 27)) * 0x94d049bb133111eb;
11    return x ^ (x >> 31);
12  }
13  size_t operator()(uint64_t x) const {
14    static const uint64_t FIXED_RANDOM = chrono::
15      steady_clock::now().time_since_epoch().count();
16    return splitmix64(x + FIXED_RANDOM);
17  }
18 };
19 gp_hash_table<int, int, custom_hash> ss;
```

3.4 PBDS Order Set [231774]

```
1 // .find_by_order(k) 回傳第 k 小的值 (based-0)
2 // .order_of_key(k) 回傳有多少元素比 k 小
3 // 不能在 #define int long long 後 #include 檔案
4 #include <ext/pb_ds/assoc_container.hpp>
5 #include <ext/pb_ds/tree_policy.hpp>
6 using namespace __gnu_pbds;
7 typedef tree<int, null_type, less<int>, rb_tree_tag,
8   tree_order_statistics_node_update> order_set;
```

3.5 Segment Tree Add Set [bb1898]

```
1 // [ll, rr), based-0
2 // 使用前記得 init(陣列大小), build(陣列名稱)
3 // add(ll, rr): 區間修改
4 // set(ll, rr): 區間賦值
5 // query(ll, rr): 區間求和 / 求最大值
6 struct SegmentTree{
7   struct node{
8     int add_tag = 0;
9     int set_tag = 0;
10    int sum = 0;
11    int ma = 0;
12  };
13  vector<node> arr;
14  SegmentTree(int n){
15    arr.resize(n<<2);
16  }
17  node pull(node A, node B){
18    node C;
19    C.sum = A.sum+B.sum;
20    C.ma = max(A.ma, B.ma);
21    return C;
22  }
23  // cce0c8
```

```
28 void push(int idx, int ll, int rr){
29   if (arr[idx].set_tag!=0){
30     arr[idx].sum = (rr-ll)*arr[idx].set_tag;
31     arr[idx].ma = arr[idx].set_tag;
32     if (rr-ll>1){
33       arr[idx*2+1].add_tag = 0;
34       arr[idx*2+1].set_tag = arr[idx].set_tag;
35       arr[idx*2+2].add_tag = 0;
36       arr[idx*2+2].set_tag = arr[idx].set_tag;
37     }
38     arr[idx].set_tag = 0;
39   }
40   if (arr[idx].add_tag!=0){
41     arr[idx].sum += (rr-ll)*arr[idx].add_tag;
42     arr[idx].ma += arr[idx].add_tag;
43     if (rr-ll>1){
44       arr[idx*2+1].add_tag += arr[idx].add_tag;
45       arr[idx*2+2].add_tag += arr[idx].add_tag;
46     }
47     arr[idx].add_tag = 0;
48   }
49 }
50
51 void build(vector<int> &v, int idx = 0, int ll = 0, int
52   rr = n){
53   if (rr-ll==1){
54     arr[idx].sum = v[ll];
55     arr[idx].ma = v[ll];
56   }else{
57     int mid = (ll+rr)/2;
58     build(v, idx*2+1, ll, mid);
59     build(v, idx*2+2, mid, rr);
60     arr[idx] = pull(arr[idx*2+1], arr[idx*2+2]);
61   }
62 }
63
64 void add(int ql, int qr, int val, int idx = 0, int ll =
65   0, int rr = n){
66   push(idx, ll, rr);
67   if (rr<=ql || qr<=ll) return;
68   if (ql<=ll && rr<=qr){
69     arr[idx].add_tag += val;
70     push(idx, ll, rr);
71     return;
72   }
73   int mid = (ll+rr)/2;
74   add(ql, qr, val, idx*2+1, ll, mid);
75   add(ql, qr, val, idx*2+2, mid, rr);
76   arr[idx]=pull(arr[idx*2+1], arr[idx*2+2]);
77 }
78
79 void set(int ql, int qr, int val, int idx=0, int ll=0,
80   int rr=n){
81   push(idx, ll, rr);
82   if (rr<=ql || qr<=ll) return;
83   if (ql<=ll && rr<=qr){
84     arr[idx].add_tag = 0;
85     arr[idx].set_tag = val;
86     push(idx, ll, rr);
87     return;
88   }
89   int mid = (ll+rr)/2;
90   set(ql, qr, val, idx*2+1, ll, mid);
91   set(ql, qr, val, idx*2+2, mid, rr);
92   arr[idx] = pull(arr[idx*2+1], arr[idx*2+2]);
93 }
```

```
91
92 node query(int ql, int qr, int idx = 0, int ll = 0, int
93   rr = n){
94   push(idx, ll, rr);
95   if (rr<=ql || qr<=ll) return node();
96   if (ql<=ll && rr<=qr) return arr[idx];
97
98   int mid = (ll+rr)/2;
99   return pull(query(ql, qr, idx*2+1, ll, mid), query(ql
100     , qr, idx*2+2, mid, rr));
101 }
```

3.6 Segment Tree Li Chao Line [45b8ba]

```
1 // 全部都是 0-based
2 // 宣告: LC_Segment_Tree st(n);
3 // 函式:
4 // update({a, b}): 插入一條 y=ax+b 的全域直線
5 // query(x): 查詢所有直線在位置 x 的最小值
6 const int MAX_V = 1e6+10; // 值域最大值
7
8 struct LC_Segment_Tree{
9   struct Node{ // y = ax+b
10     int a = 0;
11     int b = INF;
12
13     int y(int x){
14       return a*x+b;
15     }
16   };
17   vector<Node> arr;
18
19   LC_Segment_Tree(int n = 0){
20     arr.resize(4*n);
21   }
22
23   void update(Node val, int idx = 0, int ll = 0, int rr =
24     MAX_V){
25     if (rr-ll==0) return;
26     if (rr-ll==1){
27       if (val.y(ll)<arr[idx].y(ll)){
28         arr[idx] = val;
29       }
30       return;
31     }
32     int mid = (ll+rr)/2;
33     if (arr[idx].a > val.a) swap(arr[idx], val); // 原本
34     // 的線斜率要比較小
35     if (arr[idx].y(mid) < val.y(mid)){ // 交點在左邊
36       update(val, idx*2+1, ll, mid);
37     }else{ // 交點在右邊
38       swap(arr[idx], val); // 在左子樹中，新線比舊線還
39       // 要好
40       update(val, idx*2+2, mid, rr);
41     }
42     return;
43   }
44
45   int query(int x, int idx = 0, int ll = 0, int rr = MAX_V)
46   {
47     if (rr-ll==0) return INF;
48     if (rr-ll==1){
```

```

46         return arr[idx].y(l1);
47     }
48
49     int mid = (l1+rr)/2;
50     if (x<mid){
51         return min(arr[idx].y(x), query(x, idx*2+1, l1,
52             mid));
53     }else{
54         return min(arr[idx].y(x), query(x, idx*2+2, mid,
55             rr));
56     }
57 }
58 };

```

3.7 Segment Tree Li Chao Segment [cb9b37]

```

1 // 全部都是 0-based
2 // 宣告：LC_Segment_Tree st(n); // n = 值域大小 · x 的範圍為 [0, n)
3 // 函式：
4 // st.update_segment({a, b}, ql, qr)：在 [ql, qr) 插入一條 y = ax+b 的線段
5 // st.query(x)：查詢所有線段在位置 x 的最小值
6
7 struct LC_Segment_Tree {
8     struct Line { // y = ax + b
9         int a = 0, b = INF;
10        int y(int x) const { return a * x + b; }
11    };
12
13    int N;
14    vector<Line> arr;
15    LC_Segment_Tree(int n) : N(n), arr(4 * n + 5) {}
16
17    void insert_line(Line val, int idx, int l, int r) {
18        if (arr[idx].a > val.a) swap(arr[idx], val);
19        if (arr[idx].a == val.a) {
20            arr[idx].b = min(arr[idx].b, val.b);
21            return;
22        }
23        if (val.y(l) >= arr[idx].y(l)) return;
24        if (val.y(r - 1) < arr[idx].y(r - 1)) {
25            arr[idx] = val;
26            return;
27        }
28        int mid = (l + r) / 2;
29        if (val.y(mid) < arr[idx].y(mid)) {
30            swap(arr[idx], val);
31            insert_line(val, 2 * idx + 2, mid, r);
32        }
33        else insert_line(val, 2 * idx + 1, l, mid);
34    }
35
36    void dfs(Line v, int ql, int qr, int idx, int l, int r) {
37        if (l >= r || r <= ql || l >= qr) return;
38        if (ql <= l && r <= qr) {
39            insert_line(v, idx, l, r);
40            return;
41        }
42        int mid = (l + r) / 2;
43        dfs(v, ql, qr, 2 * idx + 1, l, mid);
44        dfs(v, ql, qr, 2 * idx + 2, mid, r);
45    }
46
47    int q(int x, int idx, int l, int r) {

```

```

48        if (l >= r) return INF;
49        int res = arr[idx].y(x);
50        if (l + 1 == r) return res;
51        int mid = (l + r) / 2;
52        if (x < mid) return min(res, q(x, 2*idx+1, l, mid));
53        else return min(res, q(x, 2*idx+2, mid, r));
54    }
55
56    void update_segment(Line val, int ql, int qr) {
57        dfs(val, ql, qr, 0, 0, N);
58    }
59    int query(int x) { return q(x, 0, 0, N); }
60 };

```

3.8 Segment Tree Persistent [67eafb]

```

1 // Each operation adds at most (K × ceil(log2(n))) nodes
2 // Range update : K = 8, Range query : K = 4
3 // Only point update : ceil(log2(n)) + 1
4 // To optimize K, consider 懶標永久化
5 struct Persistent_Segment_Tree { // 1-based, [L, r]
6     struct Node {
7         int lc = 0, rc = 0, sum = 0, tag = 0;
8     };
9     vector<Node> tr;
10
11    Persistent_Segment_Tree() {
12        tr.push_back({}); // Node 0 is null dummy
13    }
14
15    void apply(int u, int len, int v) {
16        tr[u].sum += v * len;
17        tr[u].tag += v;
18    }
19
20    void pushup(Node& u) {
21        u.sum = tr[u.lc].sum + tr[u.rc].sum;
22    }
23
24    Node merge(const Node& a, const Node& b) const {
25        return {0, 0, a.sum + b.sum, 0};
26    }
27
28    void pushdown(int u, int l, int r) {
29        if (!tr[u].tag) return;
30        int mid = (l + r) >> 1, tag = tr[u].tag;
31        int lc = tr.size(), rc = tr.size() + 1;
32        tr.push_back(tr[tr[u].lc]); tr.push_back(tr[tr[u].rc]);
33        apply(lc, mid - l + 1, tag); apply(rc, r - mid, tag);
34        tr[u].tag = 0; tr[u].lc = lc; tr[u].rc = rc;
35    }
36
37    // Must build before use, returns root index
38    int build(int l, int r, const vector<int>& a = {}) {
39        int u = tr.size(); tr.push_back({});
40        if (l == r) {
41            tr[u].sum = (a.empty() ? 0 : a[l]);
42            return u;
43        }
44        int mid = (l + r) >> 1;
45        int lc = build(l, mid, a), rc = build(mid + 1, r, a);
46        tr[u].lc = lc; tr[u].rc = rc;
47        pushup(tr[u]);
48        return u;
49    }

```

```

50 // Returns root index
51 int update(int pre, int l, int r, int ql, int qr, int v) {
52     int u = tr.size(); tr.push_back(tr[pre]);
53     if (ql <= l && r <= qr) {
54         apply(u, r - l + 1, v);
55         return u;
56     }
57     pushdown(u, l, r);
58     int mid = (l + r) >> 1, lc = tr[u].lc, rc = tr[u].rc;
59     if (ql <= mid) lc = update(lc, l, mid, ql, qr, v);
60     if (qr > mid) rc = update(rc, mid + 1, r, ql, qr, v);
61     tr[u].lc = lc; tr[u].rc = rc;
62     pushup(tr[u]);
63     return u;
64 }
65
66 Node query(int u, int l, int r, int ql, int qr) {
67     if (ql <= l && r <= qr) return tr[u];
68     pushdown(u, l, r);
69     int mid = (l + r) >> 1, lc = tr[u].lc, rc = tr[u].rc;
70     if (qr <= mid) return query(lc, l, mid, ql, qr);
71     if (ql > mid) return query(rc, mid + 1, r, ql, qr);
72     return merge(query(lc, l, mid, ql, qr),
73         query(rc, mid + 1, r, ql, qr));
74 }
75 };
76

```

3.9 Segment Tree Zkw [69ce93]

```

1 struct SegmentTreeZkw { // 1-based index
2     using T = array<int, 2>;
3     using U = int;
4     int n, m;
5     vector<T> seg;
6     vector<U> lazy;
7     static constexpr T empty_node = {0, 0};
8     static constexpr U empty_lazy = 1;
9
10    SegmentTreeZkw(int sz) : n(sz), m(sz - 1) {
11        seg.assign(2 * n, empty_node);
12        lazy.assign(2 * n, empty_lazy);
13    }
14    void build(T* a) {
15        for (int i = 1; i <= n; ++i) {
16            seg[m + i] = a[i];
17        }
18        for (int i = m; i >= 1; --i) {
19            seg[i] = merge(seg[i << 1], seg[i << 1 | 1]);
20        }
21    }
22    void range_update(int l, int r, U val) {
23        l += m; r += m;
24        int l0 = l, r0 = r;
25        push(l0), push(r0);
26        while (l <= r) {
27            if (l & 1) modify(l++, val);
28            if (!(r & 1)) modify(r--, val);
29            l >>= 1; r >>= 1;
30        }
31        push(l0), push(r0);
32        pull(l0), pull(r0);
33    }
34    void modify(int i, U val) {
35        seg[i][1] = seg[i][1] * val % mod;

```

```

36     lazy[i] = lazy[i] * val % mod;
37 }
38 void push(int i) {
39     for (int h = __lg(i); h >= 1; h--) {
40         int idx = i >> h;
41         if (lazy[idx] == empty_lazy) continue;
42         modify(idx << 1, lazy[idx]);
43         modify(idx << 1 | 1, lazy[idx]);
44         lazy[idx] = empty_lazy;
45     }
46 }
47 void pull(int i) {
48     while (i >>= 1) {
49         seg[i] = merge(seg[i << 1], seg[i << 1 | 1]);
50     }
51 }
52 T merge(T x, T y) {
53     return {x[0] + y[0], (x[1] + hmul_pow[x[0]] * y[1]) %
54         mod};
55 }
56 T query(int l, int r) { // [l, r]
57     T resl = empty_node, resr = empty_node;
58     l += m; r += m;
59     push(l), push(r);
60     while (l <= r) {
61         if (l & 1) resl = merge(resl, seg[l++]);
62         if (!(r & 1)) resr = merge(seg[r--], resr);
63         l >>= 1; r >>= 1;
64     }
65     return merge(resl, resr);
66 };

```

3.10 Sparse Table [31f22a]

```

1 struct SparseTable{
2     vector<vector<int>> st;
3     void build(vector<int> v){
4         int h = __lg(v.size());
5         st.resize(h+1);
6         st[0] = v;
7
8         for (int i=1 ; i<=h ; i++){
9             int gap = (1<<(i-1));
10            for (int j=0 ; j+gap<st[i-1].size() ; j++){
11                st[i].push_back(min(st[i-1][j], st[i-1][j+gap]
12                    ));
13            }
14        }
15    }
16    // 回傳 [ll, rr] 的最小值
17    int query(int ll, int rr){
18        int h = __lg(rr-ll);
19        return min(st[h][ll], st[h][rr-(1<<h)]);
20    }
21 };

```

3.11 Treap2 [3b0cca]

```

1 // 1-based · 請注意 MAX_N 是否足夠大
2 int root = 0;
3 int lc[MAX_N], rc[MAX_N];
4 int pri[MAX_N], val[MAX_N];
5 int sz[MAX_N], tag[MAX_N], fa[MAX_N], total[MAX_N];

```

```

6 // tag 為不包含自己 ( 僅要給子樹 ) 的資訊
7 int nodeCnt = 0;
8 int& new_node(int v){
9     nodeCnt++;
10    val[nodeCnt] = v;
11    total[nodeCnt] = v;
12    sz[nodeCnt] = 1;
13    pri[nodeCnt] = rand();
14    return nodeCnt;
15 }
16
17 void apply(int x, int V){
18     val[x] += V;
19     tag[x] += V;
20     total[x] += V*sz[x];
21 }
22
23 void push(int x){
24     if (tag[x]){
25         if (lc[x]) apply(lc[x], tag[x]);
26         if (rc[x]) apply(rc[x], tag[x]);
27     }
28     tag[x] = 0;
29 }
30
31 int pull(int x){
32     if (x){
33         fa[x] = 0;
34         sz[x] = 1+sz[lc[x]]+sz[rc[x]];
35         total[x] = val[x]+total[lc[x]]+total[rc[x]];
36         if (lc[x]) fa[lc[x]] = x;
37         if (rc[x]) fa[rc[x]] = x;
38     }
39     return x;
40 }
41
42 int merge(int a, int b){
43     if (!a or !b) return a|b;
44     push(a), push(b);
45
46     if (pri[a]>pri[b]){
47         rc[a] = merge(rc[a], b);
48         return pull(a);
49     }else{
50         lc[b] = merge(a, lc[b]);
51         return pull(b);
52     }
53 }
54 // [1, k] [k+1, n]
55 void split(int x, int k, int &a, int &b) {
56     if (!x) return a = b = 0, void();
57     push(x);
58     if (sz[lc[x]] >= k) {
59         split(lc[x], k, a, lc[x]);
60         b = x;
61         pull(a); pull(b);
62     }else{
63         split(rc[x], k - sz[lc[x]] - 1, rc[x], b);
64         a = x;
65         pull(a); pull(b);
66     }
67 }
68
69 // functions
70 // 回傳 x 在 Treap 中的位置

```

```

71 int get_pos(int x){
72     vector<int> sta;
73     while (fa[x]){
74         sta.push_back(x);
75         x = fa[x];
76     }
77     while (sta.size()){
78         push(x);
79         x = sta.back();
80         sta.pop_back();
81     }
82     push(x);
83
84     int res = sz[x] - sz[rc[x]];
85     while (fa[x]){
86         if (rc[fa[x]]==x){
87             res += sz[fa[x]]-sz[x];
88         }
89         x = fa[x];
90     }
91     return res;
92 }
93
94 // 1-based <前 [1, l-1] 個元素, [l, r] 個元素, [r+1, n] 個元
95 // 素>
96 array<int, 3> cut(int x, int l, int r){
97     array<int, 3> ret;
98     split(x, r, ret[1], ret[2]);
99     split(ret[1], l-1, ret[0], ret[1]);
100    return ret;
101 }
102
103 void print(int x){
104     push(x);
105     if (lc[x]) print(lc[x]);
106     cerr << val[x] << " ";
107     if (rc[x]) print(rc[x]);
108 }

```

3.12 Trie [b6475c]

```

1 struct Trie{
2     struct Data{
3         int nxt[2]={0, 0};
4     };
5
6     int sz=0;
7     vector<Data> arr;
8
9     void init(int n){
10        arr.resize(n);
11    }
12
13    void insert(int n){
14        int now=0;
15        for (int i=N ; i>=0 ; i--){
16            int v=(n>>i)&1;
17            if (!arr[now].nxt[v]){
18                arr[now].nxt[v]=++sz;
19            }
20            now=arr[now].nxt[v];
21        }
22    }
23
24    int query(int n){

```

```

25     int now=0, ret=0;
26     for (int i=N ; i>=0 ; i--){
27         int v=(n>>i)&1;
28         if (arr[now].nxt[1-v]){
29             ret+=(1<<i);
30             now=arr[now].nxt[1-v];
31         }else if (arr[now].nxt[v]){
32             now=arr[now].nxt[v];
33         }else{
34             return ret;
35         }
36     }
37     return ret;
38 }
39 } tr;
40

```

4 Dynamic-Programming

4.1 Digit DP [133f00]

```

1 #include <bits/stdc++.h>
2 using namespace std;
3
4 long long l, r;
5 long long dp[20][10][2][2]; // dp[pos][pre][Limit] = 後 pos
6 // 位 · pos 前一位是 pre · (是/否) 有上界 · (是/否) 有前綴零
7 // 的答案數量
8
9 long long memorize_search(string &s, int pos, int pre, bool
10 limit, bool lead){
11
12     // 已經被找過了 · 直接回傳值
13     if (dp[pos][pre][limit][lead]!=-1) return dp[pos][pre][
14 limit][lead];
15
16     // 已經搜尋完畢 · 紀錄答案並回傳
17     if (pos==(int)s.size()){
18         return dp[pos][pre][limit][lead] = 1;
19     }
20
21     // 枚舉目前的位數數字是多少
22     long long ans = 0;
23     for (int now=0 ; now<=(limit ? s[pos]-'0' : 9) ; now++){
24         if (now==pre){
25
26             // 1~9 絕對不能連續出現
27             if (pre!=0) continue;
28
29             // 如果已經不在前綴零的範圍內 · 0 不能連續出現
30             if (lead==false) continue;
31
32             ans += memorize_search(s, pos+1, now, limit&(now==(s[
33 pos]-'0')), lead&(now==0));
34
35         }
36     }
37
38     // 已經搜尋完畢 · 紀錄答案並回傳
39     return dp[pos][pre][limit][lead] = ans;
40 }
41
42 // 回傳 [0, n] 有多少數字符合條件
43 long long find_answer(long long n){
44     memset(dp, -1, sizeof(dp));
45

```

```

39     string tmp = to_string(n);
40
41     return memorize_search(tmp, 0, 0, true, true);
42 }
43
44 int main(){
45
46     // input
47     cin >> l >> r;
48
49     // output - 計算 [l, r] 有多少數字任意兩個位數都不相同
50     cout << find_answer(r)-find_answer(l-1) << "\n";
51
52     return 0;
53 }

```

4.2 Integer Partition

$dp[i][x]$ = 要將整數 x 拆成 i 堆的「組合數」

$dp[i+1][x+1] += dp[i][x]$ (創造新的一堆)
 $dp[i][x+i] += dp[i][x]$ (把每一堆都增加1)

4.3 Knapsack On Tree [df69b1]

```

1 // 需要重構、需要增加註解
2 #include <bits/stdc++.h>
3 #define F first
4 #define S second
5 #define all(x) begin(x), end(x)
6 using namespace std;
7
8 #define chmax(a, b) (a) = (a) < (b) ? (b) : (a)
9 #define chmin(a, b) (a) = (a) < (b) ? (a) : (b)
10
11 #define ll long long
12
13 #define FOR(i, a, b) for (int i = a; i <= b; i++)
14
15 int N, W, cur;
16 vector<int> w, v, sz;
17 vector<vector<int>> adj, dp;
18
19 void dfs(int x) {
20     sz[x] = 1;
21     for (int i : adj[x]) dfs(i), sz[x] += sz[i];
22     cur++;
23     // choose x
24     for (int i=w[x] ; i<=W ; i++){
25         dp[cur][i] = dp[cur - 1][i - w[x]] + v[x];
26     }
27     // not choose x
28     for (int i=0 ; i<=W ; i++){
29         chmax(dp[cur][i], dp[cur - sz[x]][i]);
30     }
31 }
32
33 signed main() {
34     cin >> N >> W;
35     adj.resize(N + 1);
36     w.assign(N + 1, 0);
37     v.assign(N + 1, 0);
38     sz.assign(N + 1, 0);
39     dp.assign(N + 2, vector<int>(W + 1, 0));
40     for (int i=1 ; i<=N ; i++){
41         int p; cin >> p;

```

```

42         adj[p].push_back(i);
43     }
44
45     for (int i=1 ; i<=N ; i++) cin >> w[i];
46     for (int i=1 ; i<=N ; i++) cin >> v[i];
47     dfs(0);
48     cout << dp[N + 1][W] << '\n';
49 }

```

4.4 SOS DP [8dfa8b]

```

1 // 總時間複雜度為  $O(n \cdot 2^n)$ 
2 // 計算  $dp[i] = i$  所有 bit mask 子集的和
3 for (int i=0 ; i<n ; i++){
4     for (int mask=0 ; mask<(1<<n) ; mask++){
5         if ((mask>>i)&1){
6             dp[mask] += dp[mask^(1<<i)];
7         }
8     }
9 }

```

5 Geometry

5.1 misc [3ef351]

```

1 using ld = double;
2
3 // 判斷數值正負 : {1:正數,0:零,-1:負數}
4 int sign(long long x) {return (x >= 0) ? ((bool)x) : -1; }
5 int sign(ld x) {return (abs(x) < 1e-9) ? 0 : (x>0 ? 1 : -1); }

```

5.2 convex hull [bbcaaf]

```

1 int sign(long long x) {return (x >= 0) ? ((bool)x) : -1; }
2
3 template<typename T>
4 struct point {
5     T x, y;
6     point() {}
7     point(const T &x0, const T &y0) : x(x0), y(y0) {}
8
9     point operator-(point b) {return {x-b.x, y-b.y}; }
10    bool operator==(point b) {return x==b.x && y==b.y; }
11    T operator^(point b) {return x * b.y - y * b.x; }
12    friend int ori(point a, point b, point c) {
13        return sign((b-a)^(c-a));
14    }
15 };
16
17 template<typename T>
18 struct polygon {
19     vector<point<T>> v;
20     void make_convex_hull(int simple) {
21         auto cmp = [&](point<T> &p, point<T> &q) {
22             return (p.x == q.x) ? (p.y < q.y) : (p.x < q.x);
23         };
24         simple = (bool)simple;
25         sort(v.begin(), v.end(), cmp);
26         v.resize(unique(v.begin(), v.end()) - v.begin());
27         if (v.size() <= 1) return;
28         vector<point<T>> hull;
29         for (int t = 0; t < 2; ++t){
30             int sz = hull.size();
31             for (auto &i:v) {

```

```

32         while (hull.size() >= sz+2 && ori(hull[hull.
33             size()-2], hull.back(), i) < simple) {
34             hull.pop_back();
35         }
36         hull.push_back(i);
37     }
38     hull.pop_back();
39     reverse(v.begin(), v.end());
40 }
41 swap(hull, v);
42 }
43 };

```

5.3 SVG writer [5bf313]

```

1 class SVG {
2     void p(string_view s) { o << s; }
3     void p(string_view s, auto v, auto... vs) {
4         auto i = s.find('$');
5         o << s.substr(0, i) << v, p(s.substr(i + 1), vs...);
6     }
7     ofstream o;
8     string c = "red";
9 public:
10     SVG(auto f, auto x1, auto y1, auto x2, auto y2) : o(f) {
11         p("<svg xmlns='http://www.w3.org/2000/svg' "
12           "viewBox='$ $ $ $'>\n"
13           "<style>{*{stroke-width:0.5%};}</style>\n",
14           x1, -y2, x2 - x1, y2 - y1);
15     } // cc67e3
16     ~SVG() { p("</svg>\n"); }
17     void color(string nc) { c = nc; }
18     void line(auto x1, auto y1, auto x2, auto y2) {
19         p("<line x1='$' y1='$' x2='$' y2='$' stroke='$'>\n",
20           x1, -y1, x2, -y2, c);
21     } // a08d95
22     void circle(auto x, auto y, auto r) {
23         p("<circle cx='$' cy='$' r='$' stroke='$' "
24           "fill='$'>\n", x, -y, r, c, "none");
25     } // 73c400
26     void text(auto x, auto y, string s, int w = 12) {
27         p("<text x='$' y='$' font-size='$px'>${s}</text>\n",
28           x, -y, w, s);
29     } // a23944
30 };
31 // declare : SVG svg("test.svg", -99, -99, 100, 100);

```

5.4 struct point [f21f07]

```

1 template<typename T>
2 struct point {
3     T x, y;
4     point() {}
5     point(const T &x0, const T &y0) : x(x0), y(y0) {}
6     explicit operator point<ld>() {return point<ld>(x, y); }
7 };

```

5.5 point / basic operation [c415da]

```

1 point operator+(point b) {return {x+b.x, y+b.y}; }
2 point operator-(point b) {return {x-b.x, y-b.y}; }
3 point operator*(T b) {return {x*b, y*b}; }
4 point operator/(T b) {return {x/b, y/b}; }
5 bool operator==(point b) {return x==b.x && y==b.y; }

```

```

6 T operator*(point b) {return x * b.x + y * b.y; }
7 T operator^(point b) {return x * b.y - y * b.x; }

```

5.6 point / operators [fc43e4]

```

1 // 逆時針極角排序
2 bool side() const{return (y == 0) ? (x > 0) : (y < 0); }
3 bool operator<(const point &b) const {
4     return side() == b.side() ?
5         (x*b.y > b.x*y) : side() < b.side();
6 }
7 friend ostream& operator<<(ostream& os, point p) {
8     return os << "(" << p.x << ", " << p.y << ")";
9 }
10 // 判斷 ab 到 ac 的方向 : {1:逆時鐘,0:重疊,-1:順時鐘}
11 friend int ori(point a, point b, point c) {
12     return sign((b-a)^(c-a));
13 }
14 friend bool btw(point a, point b, point c) {
15     return ori(a, b, c) == 0 && sign((a-c)*(b-c)) <= 0;
16 }

```

5.7 point / banana [09fd7c]

```

1 // 判斷線段 ab, cd 是否相交
2 friend bool banana(point a, point b, point c, point d) {
3     if (btw(a, b, c) || btw(a, b, d)
4         || btw(c, d, a) || btw(c, d, b)) return true;
5     int u = ori(a, b, c) * ori(a, b, d);
6     int v = ori(c, d, a) * ori(c, d, b);
7     return u < 0 && v < 0;
8 }

```

5.8 point / rayHitSeg [db541a]

```

1 // 判斷 "射線 ab" 與 "線段 cd" 是否相交
2 friend bool rayHitSeg(point a, point b, point c, point d) {
3     if (a == b) return btw(c, d, a); // Special case
4     if (((a - b) ^ (c - d)) == 0) {
5         return btw(a, c, b) || btw(a, d, b) || banana(a, b, c
6             , d);
7     }
8     point u = b - a, v = d - c, s = c - a;
9     return sign(s ^ v) * sign(u ^ v) >= 0 && sign(s ^ u)
10        * sign(u ^ v) >= 0 && abs(s ^ u) <= abs(u ^ v);

```

5.9 point / else [f46547]

```

1 // 旋轉 Arg(b) 的角度 (小心溢位)
2 point rotate(point b){return {x*b.x-y*b.y, x*b.y+y*b.x};}
3 // 回傳極座標角度・值域 : [-π, +π]
4 friend ld Arg(point b) {
5     return (b.x != 0 || b.y != 0) ? atan2(b.y, b.x) : 0;
6 }
7 friend T abs2(point b) {return b * b; }

```

5.10 struct line [210dc8]

```

1 template<typename T>
2 struct line {

```

```

3     point<T> p1, p2;
4     // ax + by + c = 0
5     T a, b, c; // |a|, |b| ≤ 2C, |c| ≤ 8C²
6     line() {}
7     line(const point<T> &x, const point<T> &y) : p1(x), p2(y){
8         build();
9     }
10    void build() {
11        a = p1.y - p2.y;
12        b = p2.x - p1.x;
13        c = (-a*p1.x)-b*p1.y;
14    }
15 };

```

5.11 line / else [09bcc9]

```

1 // 判斷點和有向直線的關係 : {1:左邊,0:在線上,-1:右邊}
2 int ori(point<T> &p) {
3     return sign((p2-p1) ^ (p-p1));
4 }
5 // 判斷直線斜率是否相同
6 bool parallel(line &l) {
7     return ((p1-p2) ^ (l.p1-l.p2)) == 0;
8 }
9 // 兩直線交點
10 point<ld> line_intersection(line &l) {
11     using P = point<ld>;
12     point<T> u = p2-p1, v = l.p2-l.p1, s = l.p1-p1;
13     return P(p1) + P(u) * ((ld(s^v)) / (u^v));
14 }

```

5.12 struct polygon [171a85]

```

1 template<typename T>
2 struct polygon {
3     vector<point<T>> v;
4     polygon() {}
5     polygon(const vector<point<T>> &u) : v(u) {}
6 };

```

5.13 polygon / make convex hull [2bb3ef]

```

1 // simple 為 true 的時候會回傳任意三點不共線的凸包
2 void make_convex_hull(int simple) {
3     auto cmp = [&](point<T> &p, point<T> &q) {
4         return (p.x == q.x) ? (p.y < q.y) : (p.x < q.x);
5     };
6     simple = (bool)simple;
7     sort(v.begin(), v.end(), cmp);
8     v.resize(unique(v.begin(), v.end()) - v.begin());
9     if (v.size() <= 1) return;
10    vector<point<T>> hull;
11    for (int t = 0; t < 2; ++t){
12        int sz = hull.size();
13        for (auto &i:v) {
14            while (hull.size() >= sz+2 && ori(hull[hull.
15                size()-2], hull.back(), i) < simple) {
16                hull.pop_back();
17            }
18            hull.push_back(i);
19        }
20        hull.pop_back();
21        reverse(v.begin(), v.end());
22    }

```

```
23 swap(hull, v);
24 }
```

5.14 polygon / in polygon [f11340]

```
1 // 可以在有  $n$  個點的簡單多邊形內，用  $O(n)$  判斷一個點：
2 // {1 : 在多邊形內, 0 : 在多邊形上, -1 : 在多邊形外}
3 int in_polygon(point<T> a){
4     const T MAX_POS = 1e9 + 5; // [記得修改] 座標的最大值
5     point<T> pre = v.back(), b(MAX_POS, a.y + 1);
6     int cnt = 0;
7
8     for (auto &i:v) {
9         if (btw(pre, i, a)) return 0;
10        if (banana(a, b, pre, i)) cnt++;
11        pre = i;
12    }
13
14    return cnt%2 ? 1 : -1;
15 }
```

5.15 polygon / in convex [e64f1e]

```
1 /// 警告：以下所有凸包專用的函式都只接受逆時針排序且任三點不
2   共線的凸包 ///
3 // 可以在有  $n$  個點的凸包內，用  $O(\log n)$  判斷一個點：
4 // {1 : 在凸包內, 0 : 在凸包邊上, -1 : 在凸包外}
5 int in_convex(point<T> p) {
6     int n = v.size();
7     int a = ori(v[0], v[1], p), b = ori(v[0], v[n-1], p);
8     if (a < 0 || b > 0) return -1;
9     if (btw(v[0], v[1], p)) return 0;
10    if (btw(v[0], v[n-1], p)) return 0;
11    int l = 1, r = n - 1, mid;
12    while (l + 1 < r) {
13        mid = (l + r) >> 1;
14        if (ori(v[0], v[mid], p) >= 0) l = mid;
15        else r = mid;
16    }
17    int k = ori(v[l], v[r], p);
18    if (k <= 0) return k;
19    return 1;
20 }
```

5.16 polygon / cycle search [fe2f51]

```
1 // 凸包專用的環狀二分搜，回傳  $\theta$ -based index
2 int cycle_search(auto &f) {
3     int n = v.size(), l = 0, r = n;
4     if (n == 1) return 0;
5     bool rv = f(1, 0);
6     while (r - l > 1) {
7         int m = (l + r) / 2;
8         if (f(0, m) ? rv : f(m, (m + 1) % n)) r = m;
9         else l = m;
10    }
11    return f(1, r % n) ? 1 : r % n;
12 }
```

5.17 polygon / line cut convex [b6a4c8]

```
1 // 可以在有  $n$  個點的凸包內，用  $O(\log n)$  判斷一條直線：
```

```
2 // {1 : 穿過凸包, 0 : 剛好切過凸包, -1 : 沒碰到凸包}
3 int line_cut_convex(line<T> L) {
4     L.build();
5     point<T> p(L.a, L.b);
6     auto gt = [&](int neg) {
7         auto f = [&](int x, int y) {
8             return sign((v[x] - v[y]) * p) == neg;
9         };
10        return -(v[cycle_search(f)] * p);
11    };
12    T x = gt(1), y = gt(-1);
13    if (L.c < x || y < L.c) return -1;
14    return not (L.c == x || L.c == y);
15 }
```

5.18 polygon / convex line intersect [47e699]

```
1 // return edge endpoint index
2 // but if (first == second) => intersect at a vertex
3 vector<pair<int,int>> convex_line_intersect(line<T> L) {
4     auto dis = [&](int p){
5         return (L.p2 - L.p1) ^ (v[p] - L.p1);
6     };
7     auto gao = [&](int s) {
8         auto f = [&](int i, int j) {
9             return sign(dis(i) - dis(j)) == s;
10        };
11        return cycle_search(f);
12    };
13    int x = gao(1), y = gao(-1), n = v.size();
14    if (sign(dis(x)) < 0 || sign(dis(y)) > 0) return {};
15    if (sign(dis(x)) == 0 || sign(dis(y)) == 0) {
16        int j = ((sign(dis(x)) == 0 ? x : y) + n - 1) % n;
17        vector<pair<int,int>> ret;
18        for (int i = 0; i < 3; ++i, j = (j + 1) % n)
19            if (sign(dis(j)) == 0) ret.emplace_back(j, j);
20        return ret;
21    }
22    auto g = [&](int l, int r, int s) -> pair<int,int> {
23        while ((l + 1) % n != r) {
24            int m = ((l + r + (l < r ? 0 : n)) / 2) % n;
25            if (sign(dis(m)) == s ? l : r) = m;
26        }
27        return {sign(dis(r)) ? l : r, r};
28    };
29    return {g(x, y, 1), g(y, x, -1)};
30 }
```

5.19 polygon / segment pass convex interior [7a3f22]

```
1 // 可以在有  $n$  個點的凸包內，用  $O(\log n)$  判斷一個線段：
2 // {1 : 線段上存在某一點位於凸包內部（邊上不算），
3 // 0 : 線段上存在某一點碰到凸包的邊但線段上任一點均不在凸包
4 // -1 : 線段完全在凸包外面}
5 int segment_pass_convex_interior(line<T> L) {
6     if (in_convex(L.p1) == 1 || in_convex(L.p2) == 1) return
7         1;
8     L.build();
9     auto res = convex_line_intersect(L);
10    if (res.empty()) return -1;
11    bool cntp = 0, cntn = 0;
12    for (auto &[i, j] : res) if (banana(v[i], v[j], L.p1, L.p2)) {
```

```
12         int now = L.ori(v[(i + 1) % v.size()]);
13         cntp |= (now == 1);
14         cntn |= (now == -1);
15     }
16    return (-1 + cntp + cntn);
17 }
```

5.20 polygon / convex tangent point [a6f66b]

```
1 // 回傳點過凸包的兩條切線的切點的  $\theta$ -based index (不保證兩條
2   切線的順逆時針關係)
3 pair<int,int> convex_tangent_point(point<T> p) {
4     int n = v.size(), z = -1, edg = -1;
5     auto gt = [&](int neg) {
6         auto check = [&](int x) {
7             if (v[x] == p) z = x;
8             if (btw(v[x], v[(x + 1) % n], p)) edg = x;
9             if (btw(v[(x + n - 1) % n], v[x], p)) edg = (x +
10                 n - 1) % n;
11         };
12         auto f = [&](int x, int y) {
13             check(x); check(y);
14             return ori(p, v[x], v[y]) == neg;
15         };
16         return cycle_search(f);
17     };
18    int x = gt(1), y = gt(-1);
19    if (z != -1) {
20        return {(z + n - 1) % n, (z + 1) % n};
21    }
22    else if (edg != -1) {
23        return {edg, (edg + 1) % n};
24    }
25    else {
26        return {x, y};
27    }
28 }
```

5.21 polygon / halfplane intersection [102d48]

```
1 friend int halfplane_intersection(vector<line<T>> &s, polygon
2   <T> &P) {
3     auto angle_cmp = [&](line<T> &A, line<T> &B) {
4         point<T> a = A.p2-A.p1, b = B.p2-B.p1;
5         return (a < b);
6     };
7     sort(s.begin(), s.end(), angle_cmp); // 線段左側為該線段
8     int L, R, n = s.size();
9     vector<point<T>> px(n);
10    vector<line<T>> q(n);
11    q[L = R = 0] = s[0];
12    for(int i = 1; i < n; ++i) {
13        while(L < R && s[i].ori(px[R-1]) <= 0) --R;
14        while(L < R && s[i].ori(px[L]) <= 0) ++L;
15        q[++R] = s[i];
16        if(q[R].parallel(q[R-1])) {
17            --R;
18            if(q[R].ori(s[i].p1) > 0) q[R] = s[i];
19        }
20        if(L < R) px[R-1] = q[R-1].line_intersection(q[R]);
21    }
22    while(L < R && q[L].ori(px[R-1]) <= 0) --R;
23    P.v.clear();
```

```

23 |     if(R - L <= 1) return 0;
24 |     px[R] = q[R].line_intersection(q[L]);
25 |     for(int i = L; i <= R; ++i) P.v.push_back(px[i]);
26 |     return R - L + 1;
27 | }

```

5.22 polygon / minkowski sum [aba519]

```

1 | void minkowski(vector<point<T>> a, vector<point<T>> b) {
2 |     // a, b: convex polygon's inside vector, CCW order
3 |     auto cmp = [&](auto &a, auto &b) {
4 |         return tie(a.y, a.x) < tie(b.y, b.x);
5 |     };
6 |     auto reorder = [&](auto &u) {
7 |         auto it = min_element(u.begin(), u.end(), cmp);
8 |         rotate(u.begin(), it, u.end());
9 |     };
10 |    reorder(a); reorder(b);
11 |    a.push_back(a[0]); a.push_back(a[1]);
12 |    b.push_back(b[0]); b.push_back(b[1]);
13 |    v.clear();
14 |    for (int i = 0, j = 0; i+2<a.size()||j+2<b.size(); ) {
15 |        v.push_back(a[i] + b[j]);
16 |        auto val = (a[i + 1] - a[i]) ^ (b[j + 1] - b[j]);
17 |        if (val >= 0) i++;
18 |        if (val <= 0) j++;
19 |    }
20 | }

```

5.23 struct Cir [87198d]

```

1 | struct Cir {
2 |     point<ld> o; ld r;
3 |     friend ostream& operator<<(ostream& os, Cir c) {
4 |         return os << "(" << "x" << "+" << [c.o.x >= 0] << abs(c.o.x)
5 |             << ")^2 + (y" << "+" << [c.o.y >= 0] << abs(c.o.y)
6 |             << ")^2 = " << c.r * c.r;
7 |     }
8 |     bool covers(Cir b) {
9 |         return sqrt((ld)abs2(o - b.o)) + b.r <= r;
10 |    }
11 | };

```

5.24 Cir / Cir intersect [330a1c]

```

1 | vector<point<ld>> Cir_intersect(Cir c) {
2 |     ld d2 = abs2(o - c.o), d = sqrt(d2);
3 |     if (d < max(r, c.r) - min(r, c.r) || d > r + c.r) return {};
4 |     auto sqdf = [&](ld x, ld y) { return x*x - y*y; };
5 |     point<ld> u = (o + c.o) / 2 + (o - c.o) * (sqdf(c.r, r) /
6 |         (2 * d2));
7 |     ld A = sqrt(sqdf(r + d, c.r) * sqdf(c.r, d - r));
8 |     point<ld> v = (c.o - o).rotate({0,1}) * A / (2 * d2);
9 |     if (sign(v.x) == 0 && sign(v.y) == 0) return {u};
10 |    return {u - v, u + v};

```

5.25 Cir / point tangent [0067e6]

```

1 | auto point_tangent(point<ld> p) {
2 |     vector<point<ld>> res;
3 |     ld d_sq = abs2(p - o);
4 |     if (sign(d_sq - r * r) <= 0) {

```

```

5 |         res.pb(p + (p - o).rotate({0, 1}));
6 |     } else if (d_sq > r * r) {
7 |         ld s = d_sq - r * r;
8 |         point<ld> v = p + (o - p) * s / d_sq;
9 |         point<ld> u = (o - p).rotate({0, 1}) * sqrt(s) * r /
10 |             d_sq;
11 |         res.pb(v + u);
12 |         res.pb(v - u);
13 |     }
14 |     return res;

```

5.26 Cir / circle tangent [a3c7fb]

```

1 | // returns nothing when two circles are the same
2 | auto circle_tangent(Cir c2, int sign1) {
3 |     // sign1 = 1 for outer tang, -1 for inter tang
4 |     using ptld = point<ld>;
5 |     vector<pair<ptld, ptld>> res;
6 |     ld d_sq = abs2(o - c2.o);
7 |     if (sign(d_sq) == 0) return res;
8 |     ld d = sqrt(d_sq);
9 |     ptld v = (c2.o - o) / d;
10 |    ld c = (r - sign1 * c2.r) / d;
11 |    if (c * c > 1) return res;
12 |    ld h = sqrt(max((ld)0.0, 1.0 - c * c));
13 |    for (int sign2 = 1; sign2 >= -1; sign2 -= 2) {
14 |        ptld n(v.x * c - sign2 * h * v.y, v.y * c + sign2 * h
15 |            * v.x);
16 |        ptld p1 = o + n * r;
17 |        ptld p2 = c2.o + n * (c2.r * sign1);
18 |        if (sign(p1.x - p2.x) == 0 && sign(p1.y - p2.y) == 0)
19 |            p2 = p1 + (c2.o - o).rotate({0, 1});
20 |        res.push_back({p1, p2});
21 |    }
22 |    if (sign1 == -1 && sign(r + c2.r - d) == 0) res.pop_back();
23 |    if (sign1 == 1 && sign(abs(r - c2.r) - d) == 0) res.
24 |        pop_back();
25 |    return res;

```

5.27 Cir / minimum enclosing circle [3e1440]

```

1 | point<ld> circenter(point<ld> p0, point<ld> p1, point<ld> p2) {
2 |     // radius = abs(center)
3 |     p1 = p1 - p0, p2 = p2 - p0;
4 |     ld x1 = p1.x, y1 = p1.y, x2 = p2.x, y2 = p2.y;
5 |     ld m = 2. * (x1 * y2 - y1 * x2);
6 |     point<ld> center(0, 0);
7 |     center.x = (x1 * x1 * y2 - x2 * x2 * y1 + y1 * y2 * (y1 -
8 |         y2)) / m;
9 |     center.y = (x1 * x2 * (x2 - x1) - y1 * y1 * x2 + x1 * y2
10 |         * y2) / m;
11 |    return center + p0;
12 | }
13 | void min_enclosing(vector<point<ld>> p) {
14 |     shuffle(p.begin(), p.end(), rngf);
15 |     r = 0.0;
16 |     point<ld> cent = p[0];
17 |     for (int i = 1; i < p.size(); ++i) {
18 |         if (abs2(cent - p[i]) <= r) continue;
19 |         cent = p[i], r = 0.0;
20 |         for (int j = 0; j < i; ++j) {

```

```

21 |             if (abs2(cent - p[j]) <= r) continue;
22 |             cent = (p[i] + p[j]) / 2, r = abs2(p[j] - cent);
23 |             for (int k = 0; k < j; ++k) {
24 |                 if (abs2(cent - p[k]) <= r) continue;
25 |                 cent = circenter(p[i], p[j], p[k]);
26 |                 r = abs2(p[k] - cent);
27 |             }
28 |         }
29 |         o = cent;
30 |         r = sqrt(r);

```

5.28 Geometry Struct 3D [4a50c9]

```

1 | using ld = long double;
2 |
3 | // 判斷數值正負：{1:正數,0:零,-1:負數}
4 | int sign(long long x) {return (x >= 0) ? ((bool)x) : -1; }
5 | int sign(ld x) {return (abs(x) < 1e-9) ? 0 : (x>0 ? 1 : -1); }
6 |
7 | template<typename T>
8 | struct pt3 {
9 |     T x, y, z;
10 |    pt3(){}
11 |    pt3(const T &x, const T &y, const T &z):x(x),y(y),z(z){}
12 |    explicit operator pt3<ld>() {return pt3<ld>(x, y, z); }
13 |
14 |    pt3 operator+(pt3 b) {return {x+b.x, y+b.y, z+b.z}; }
15 |    pt3 operator-(pt3 b) {return {x-b.x, y-b.y, z-b.z}; }
16 |    pt3 operator*(T b) {return {x * b, y * b, z * b}; }
17 |    pt3 operator/(T b) {return {x / b, y / b, z / b}; }
18 |    bool operator==(pt3 b){return x==b.x&&y==b.y&&z==b.z; }
19 |
20 |    T operator*(pt3 b) {return x*b.x+y*b.y+z*b.z; }
21 |    pt3 operator^(pt3 b) {
22 |        return pt3(y * b.z - z * b.y,
23 |            z * b.x - x * b.z,
24 |            x * b.y - y * b.x);
25 |    }
26 |    friend T abs2(pt3 b) {return b * b; }
27 |    friend T len (pt3 b) {return sqrt(abs2(b)); }
28 |    friend ostream& operator<<(ostream& os, pt3 p) {
29 |        return os << "(" << p.x << ", " <<
30 |            p.y << ", " << p.z << ")";
31 |    }
32 | };
33 |
34 | template<typename T>
35 | struct face {
36 |     int a, b, c; // 三角形在 vector 裡面的 index
37 |     pt3<T> q; // 面積向量 (朝外)
38 | };
39 |
40 | /// 警告；v 在過程中可能被修改，回傳的 face 以修改後的為準
41 | /// O(n^2) · 最多只有 2n-4 個面
42 | /// 當凸包退化時會回傳空的凸包，否則回傳凸包上的每個面
43 | template<typename T>
44 | vector<face<T>> hull3(vector<pt3<T>> &v) {
45 |     int n = v.size();
46 |     if (n < 3) return {};
47 |     // don't use "==" when you use ld
48 |     sort(all(v), [&](pt3<T> &p, pt3<T> &q) {
49 |         return sign(p.x - q.x) ? (p.x < q.x) :

```

```

50     (sign(p.y - q.y) ? p.y < q.y : p.z < q.z);
51 });
52 v.resize(unique(v.begin(), v.end()) - v.begin());
53 for (int i = 2; i <= n; ++i) {
54     if (i == n) return {};
55     if (sign(len(((v[1] - v[0]) ^ (v[i] - v[0]))))) {
56         swap(v[2], v[i]);
57         break;
58     }
59 }
60 pt3<T> tmp_q = (v[1] - v[0]) ^ (v[2] - v[0]);
61 for (int i = 3; i <= n; ++i) {
62     if (i == n) return {};
63     if (sign((v[i] - v[0]) * tmp_q)) {
64         swap(v[3], v[i]);
65         break;
66     }
67 }
68
69 vector<face<T>> f;
70 vector<vector<int>> dead(n, vector<int>(n, true));
71 auto add_face = [&](int a, int b, int c) {
72     f.emplace_back(a, b, c, (v[b] - v[a]) ^ (v[c] - v[a]));
73     dead[a][b] = dead[b][c] = dead[c][a] = false;
74 };
75 add_face(0, 1, 2);
76 add_face(0, 2, 1);
77
78 for (int i = 3; i < n; ++i) {
79     vector<face<T>> f2;
80     for (auto &[a, b, c, q] : f) {
81         if (sign((v[i] - v[a]) * q) > 0)
82             dead[a][b] = dead[b][c] = dead[c][a] = true;
83         else f2.emplace_back(a, b, c, q);
84     }
85     f.clear();
86     for (face<T> &F : f2) {
87         int arr[3] = {F.a, F.b, F.c};
88         for (int j = 0; j < 3; ++j) {
89             int a = arr[j], b = arr[(j + 1) % 3];
90             if (dead[b][a]) add_face(b, a, i);
91         }
92     }
93     f.insert(f.end(), all(f2));
94 }
95 return f;
96 } // 15ef50

```

5.29 Pick's Theorem

給定頂點坐標均是整點的簡單多邊形：面積 = 內部格點數 + 邊上格點數/2 - 1

6 Graph

6.1 SAT [5a6317]

```

1 struct TWO_SAT {
2     int n, N;
3     vector<vector<int>> G, rev_G;
4     deque<bool> used;
5     vector<int> order, comp;
6     deque<bool> assignment;
7     void init(int _n) {
8         n = _n;
9         N = _n * 2;

```

```

10         G.resize(N + 5);
11         rev_G.resize(N + 5);
12     }
13     void dfs1(int v) {
14         used[v] = true;
15         for (int u : G[v]) {
16             if (!used[u])
17                 dfs1(u);
18         }
19         order.push_back(v);
20     }
21     void dfs2(int v, int c1) {
22         comp[v] = c1;
23         for (int u : rev_G[v]) {
24             if (comp[u] == -1)
25                 dfs2(u, c1);
26         }
27     }
28     bool solve() {
29         order.clear();
30         used.assign(N, false);
31         for (int i = 0; i < N; ++i) {
32             if (!used[i])
33                 dfs1(i);
34         }
35         comp.assign(N, -1);
36         for (int i = 0, j = 0; i < N; ++i) {
37             int v = order[N - i - 1];
38             if (comp[v] == -1)
39                 dfs2(v, j++);
40         }
41         assignment.assign(n, false);
42         for (int i = 0; i < N; i += 2) {
43             if (comp[i] == comp[i + 1])
44                 return false;
45             assignment[i / 2] = (comp[i] > comp[i + 1]);
46         }
47         return true;
48     }
49     // A or B 都是 0-based
50     void add_disjunction(int a, bool na, int b, bool nb) {
51         // na is true => ~a, na is false => a
52         // nb is true => ~b, nb is false => b
53         a = 2 * a ^ na;
54         b = 2 * b ^ nb;
55         int neg_a = a ^ 1;
56         int neg_b = b ^ 1;
57         G[neg_a].push_back(b);
58         G[neg_b].push_back(a);
59         rev_G[b].push_back(neg_a);
60         rev_G[a].push_back(neg_b);
61         return;
62     }
63     void get_result(vector<int>& res) {
64         res.clear();
65         for (int i = 0; i < n; ++i)
66             res.push_back(assignment[i]);
67     }
68 };

```

6.2 Augment Path [f8a5dd]

```

1 struct AugmentPath{
2     int n, m;
3     vector<vector<int>> G;

```

```

4     vector<int> mx, my;
5     vector<int> visx, visy;
6     int stamp;
7
8     AugmentPath(int _n, int _m) : n(_n), m(_m), G(n), mx(n,
9         -1), my(m, -1), visx(n), visy(n){
10         stamp = 0;
11     }
12
13     void add(int x, int y){
14         G[x].push_back(y);
15     }
16
17     // bb03e2
18     bool dfs1(int now){
19         visx[now] = stamp;
20
21         for (auto x : G[now]){
22             if (my[x]==-1){
23                 mx[now] = x;
24                 my[x] = now;
25                 return true;
26             }
27         }
28         for (auto x : G[now]){
29             if (visx[my[x]]!=stamp && dfs1(my[x])){
30                 mx[now] = x;
31                 my[x] = now;
32                 return true;
33             }
34         }
35         return false;
36     }
37
38     vector<pair<int, int>> find_max_matching(){
39         vector<pair<int, int>> ret;
40
41         while (true){
42             stamp++;
43             int tmp = 0;
44             for (int i=0 ; i<n ; i++){
45                 if (mx[i]==-1 && dfs1(i)) tmp++;
46             }
47             if (tmp==0) break;
48         }
49         for (int i=0 ; i<n ; i++){
50             if (mx[i]!=-1){
51                 ret.push_back({i, mx[i]});
52             }
53         }
54         return ret;
55     }
56
57     // 645577
58     void dfs2(int now){
59         visx[now] = true;
60
61         for (auto x : G[now]){
62             if (my[x]!=-1 && visy[x]==false){
63                 visy[x] = true;
64                 dfs2(my[x]);
65             }
66         }
67     }
68 }

```

```

69 // 要先執行 find_max_matching 一次
70 vector<pair<int, int>> find_min_vertex_cover(){
71     fill(visx.begin(), visx.end(), false);
72     fill(visy.begin(), visy.end(), false);
73
74     vector<pair<int, int>> ret;
75     for (int i=0; i<n; i++){
76         if (mx[i]==-1) dfs2(i);
77     }
78
79     for (int i=0; i<n; i++){
80         if (visx[i]==false) ret.push_back({1, i});
81     }
82     for (int i=0; i<m; i++){
83         if (visy[i]==true) ret.push_back({2, i});
84     }
85     return ret;
86 }
87 }
88 };

```

6.3 C3C4 [875ea3]

```

1 // 0-based
2 void C3C4(vector<int> deg, vector<array<int, 2>> edges){
3     int N = deg.size();
4     int M = edges.size();
5
6     vector<int> ord(N), rk(N);
7     iota(ord.begin(), ord.end(), 0);
8     sort(ord.begin(), ord.end(), [&](int x, int y) { return
9         deg[x] > deg[y]; });
10    for (int i=0; i<N; i++) rk[ord[i]] = i;
11
12    vector<vector<int>> D(N), adj(N);
13    for (auto [u, v] : edges) {
14        if (rk[u] > rk[v]) swap(u, v);
15        D[u].emplace_back(v);
16        adj[u].emplace_back(v);
17        adj[v].emplace_back(u);
18    }
19
20    vector<int> vis(N);
21
22    int c3 = 0, c4 = 0;
23    for (int x : ord) { // c3
24        for (int y : D[x]) vis[y] = 1;
25        for (int y : D[x]) for (int z : D[y]){
26            c3 += vis[z]; // xyz is C3
27        }
28        for (int y : D[x]) vis[y] = 0;
29    }
30    for (int x : ord) { // c4
31        for (int y : D[x]) for (int z : adj[y])
32            if (rk[z] > rk[x]) c4 += vis[z]++;
33        for (int y : D[x]) for (int z : adj[y])
34            if (rk[z] > rk[x]) --vis[z];
35    } // both are O(M*sqrt(M)), test @ 2022 CCPC guangzhou
36    cout << c4 << "\n";
37 }

```

6.4 Dinic [961b34]

```

1 // 一般圖: O(EV^2)
2 // 二分圖: O(EV)

```

```

3 struct Flow{
4     using T = int; // 可以換成別的类型
5     struct Edge{
6         int v; T rc; int rid;
7     };
8     vector<vector<Edge>> G;
9     void add(int u, int v, T c){
10         G[u].push_back({v, c, G[v].size()});
11         G[v].push_back({u, 0, G[u].size()-1});
12     }
13     vector<int> dis, it;
14
15     Flow(int n){
16         G.resize(n);
17         dis.resize(n);
18         it.resize(n);
19     }
20
21     // ce56d6
22     T dfs(int u, int t, T f){
23         if (u == t || f == 0) return f;
24         for (int &i=it[u]; i<G[u].size(); i++){
25             auto &[v, rc, rid] = G[u][i];
26             if (dis[v]!=dis[u]+1) continue;
27             T df = dfs(v, t, min(f, rc));
28             if (df <= 0) continue;
29             rc -= df;
30             G[v][rid].rc += df;
31             return df;
32         }
33         return 0;
34     }
35
36     // e22e39
37     T flow(int s, int t){
38         T ans = 0;
39         while (true){
40             fill(dis.begin(), dis.end(), INF);
41             queue<int> q;
42             q.push(s);
43             dis[s] = 0;
44
45             while (q.size()){
46                 int u = q.front(); q.pop();
47                 for (auto [v, rc, rid] : G[u]){
48                     if (rc <= 0 || dis[v] < INF) continue;
49                     dis[v] = dis[u] + 1;
50                     q.push(v);
51                 }
52             }
53             if (dis[t]==INF) break;
54
55             fill(it.begin(), it.end(), 0);
56             while (true){
57                 T df = dfs(s, t, INF);
58                 if (df <= 0) break;
59                 ans += df;
60             }
61         }
62         return ans;
63     }
64
65     // the code below constructs minimum cut
66     void dfs_mincut(int now, vector<bool> &vis){
67         vis[now] = true;
68     }
69 }

```

```

68     for (auto &[v, rc, rid] : G[now]){
69         if (vis[v] == false && rc > 0){
70             dfs_mincut(v, vis);
71         }
72     }
73 }
74
75 vector<pair<int, int>> construct(int n, int s, vector<
76     pair<int, int>> &E){
77     // E is G without capacity
78     vector<bool> vis(n);
79     dfs_mincut(s, vis);
80     vector<pair<int, int>> ret;
81     for (auto &[u, v] : E){
82         if (vis[u] == true && vis[v] == false){
83             ret.emplace_back(u, v);
84         }
85     }
86     return ret;
87 }

```

6.5 Dominator Tree [52b249]

```

1 // 全部都是 0-based
2 // G 要是向有向無權圖
3 // 一開始要初始化 G(N, root)
4 // 用完之後要 build
5 // G[i] = 從 root 走到 i 時，一定要走到的點且離 i 最近
6 struct DominatorTree{
7     int N;
8     vector<vector<int>> G;
9     vector<vector<int>> buckets, rg;
10    // dfn[x] = the DFS order of x
11    // rev[x] = the vertex with DFS order x
12    // par[x] = the parent of x
13    vector<int> dfn, rev, par;
14    vector<int> sdom, dom, idom;
15    vector<int> fa, val;
16    int stamp;
17    int root;
18
19    int operator [] (int x){
20        return idom[x];
21    }
22
23    DominatorTree(int _N, int _root) :
24        N(_N),
25        G(N), buckets(N), rg(N),
26        dfn(N, -1), rev(N, -1), par(N, -1),
27        sdom(N, -1), dom(N, -1), idom(N, -1),
28        fa(N, -1), val(N, -1)
29    {
30        stamp = 0;
31        root = _root;
32    }
33
34    void add_edge(int u, int v){
35        G[u].push_back(v);
36    }
37
38    void dfs(int x){
39        rev[dfn[x] = stamp] = x;
40        fa[stamp] = sdom[stamp] = val[stamp] = stamp;
41        stamp++;
42    }

```

```

42     for (int u : G[x]){
43         if (dfn[u]==-1){
44             dfs(u);
45             par[dfn[u]] = dfn[x];
46         }
47         rg[dfn[u]].push_back(dfn[x]);
48     }
49 }
50
51 int eval(int x, bool first){
52     if (fa[x]==x) return !first ? -1 : x;
53     int p = eval(fa[x], false);
54
55     if (p==-1) return x;
56     if (sdom[val[x]]>sdom[val[fa[x]]]) val[x] = val[fa[x]
57         ];
58     fa[x] = p;
59
60     return !first ? p : val[x];
61 }
62
63 void link(int x, int y){
64     fa[x] = y;
65 }
66
67 void build(){
68     dfs(root);
69
70     for (int x=stamp-1 ; x>=0 ; x--){
71         for (int y : rg[x]){
72             sdom[x] = min(sdom[x], sdom[eval(y, true)]);
73         }
74         if (x>0) buckets[sdom[x]].push_back(x);
75         for (int u : buckets[x]){
76             int p = eval(u, true);
77             if (sdom[p]==x) dom[u] = x;
78             else dom[u] = p;
79         }
80         if (x>0) link(x, par[x]);
81     }
82
83     idom[root] = root;
84     for (int x=1 ; x<stamp ; x++){
85         if (sdom[x]!=dom[x]) dom[x] = dom[dom[x]];
86     }
87     for (int i=1 ; i<stamp ; i++) idom[rev[i]] = rev[dom[
88         i]];
89 };

```

6.6 EdgeBCC [d09eb1]

```

1 // d09eb1
2 // 0-based · 支援重邊
3 struct EdgeBCC{
4     int n, m, dep, sz;
5     vector<vector<pair<int, int>>> G;
6     vector<vector<int>> bcc;
7     vector<int> dfn, low, stk, isBridge, bccId;
8     vector<pair<int, int>> edge, bridge;
9
10    EdgeBCC(int _n) : n(_n), m(0), sz(0), dfn(n), low(n), G(n
11        ), bcc(n), bccId(n) {}

```

```

12 void add_edge(int u, int v) {
13     edge.push_back({u, v});
14     G[u].push_back({v, m});
15     G[v].push_back({u, m++});
16 }
17
18 void dfs(int now, int pre) {
19     dfn[now] = low[now] = ++dep;
20     stk.push_back(now);
21
22     for (auto [x, id] : G[now]){
23         if (!dfn[x]){
24             dfs(x, id);
25             low[now] = min(low[now], low[x]);
26         }else if (id!=pre){
27             low[now] = min(low[now], dfn[x]);
28         }
29     }
30
31     if (low[now]==dfn[now]){
32         if (pre!=-1) isBridge[pre] = true;
33         int u;
34         do{
35             u = stk.back();
36             stk.pop_back();
37             bcc[sz].push_back(u);
38             bccId[u] = sz;
39         } while (u!=now);
40         sz++;
41     }
42 }
43
44 void get_bcc() {
45     isBridge.assign(m, 0);
46     dep = 0;
47     for (int i=0 ; i<n ; i++){
48         if (!dfn[i]) dfs(i, -1);
49     }
50
51     for (int i=0 ; i<m ; i++){
52         if (isBridge[i]){
53             bridge.push_back({edge[i].first , edge[i].
54                 second});
55         }
56     }
57 }

```

6.7 EnumeratePlanarFace [8df3e2]

```

1 struct PlanarGraph { // 0-based index
2     using T = int;
3     int n, m, faces;
4     vector<point<T>> v;
5     vector<vector<pair<int, int>>> D, G;
6     vector<vector<int>> F;
7     /* D : 對偶圖 · 平面圖的邊的編號是 [0, m - 1],
8        outer face 的編號是 m, face 的編號是 [m + 1, m + faces]
9        F : 存每個 inner face 的 point index */
10    vector<int> conv, nxt, vis;
11
12    PlanarGraph(int n, vector<point<T>> _v) :
13        n(n), m(0), faces(0), v(_v), G(n) {}
14
15    void add_edge(int x, int y) {

```

```

16        G[x].emplace_back(y, 2 * m);
17        G[y].emplace_back(x, 2 * m + 1);
18        conv.push_back(x);
19        conv.push_back(y);
20        m++;
21    }
22
23    vector<T> enumerate_face() {
24        D.resize(m + 1);
25        F.resize(m + 1);
26        nxt.resize(2 * m);
27        vis.resize(2 * m);
28        for (int i = 0; i < n; i++) {
29            sort(G[i].begin(), G[i].end(), [&](auto& a,
30                auto& b) {
31                    return (v[a.first]-v[i]) < (v[b.first]-v[i]);
32                }); // 極角排序用的 '<' operator
33        int sz = G[i].size(), pre = sz - 1;
34        for (int j = 0; j < sz; pre = j, j++) {
35            nxt[G[i][pre].second] = G[i][j].second ^ 1;
36        }
37    }
38
39    vector<T> ret;
40    // 這個 for 迴圈的 hash value 是 fa2180
41    for (int i = 0; i < 2 * m; i++) if (vis[i] == false){
42        vector<int> pt, outdeg;
43        for (int now = i; !vis[now]; now = nxt[now]) {
44            vis[now] = true;
45            pt.push_back(conv[now]);
46            outdeg.push_back(now / 2);
47        }
48
49        T area = -(v[pt.back()] ^ v[pt.front()]);
50        for (int j = 0; j + 1 < pt.size(); j++) {
51            area -= (v[pt[j]] ^ v[pt[j + 1]]);
52        }
53
54        if (area > 0) { // inner face
55            F.push_back(move(pt));
56            ret.push_back(area);
57            faces += 1;
58            D.emplace_back();
59            for (auto &u:outdeg) {
60                D[m + faces].emplace_back(u, 0);
61                D[u].emplace_back(m + faces, 1);
62            }
63        }
64        else { // outer face or a tree
65            for (auto &u:outdeg) {
66                D[m].emplace_back(u, 0);
67            }
68        }
69    }
70    return ret;
71 }
72 };

```

6.8 Ford Fulkerson [5bc854]

```

1 const int maxn = 1e5 + 10, INF = 1e9;
2 const long long INF64 = 1e18;
3 struct edge{ int to, cap, rev; };
4 vector<edge> G[maxn];
5 int n, m, s, t, a, b, c;

```

```

6 bool vis[maxn];
7 int dfs(int v, int t, int f) {
8     // cout << v << ' ' << t << ' ' << f << '\n';
9     if (v == t) return f;
10    vis[v] = true;
11    for (edge &e: G[v]) {
12        if (!vis[e.to] && e.cap > 0) {
13            int d = dfs(e.to, t, min(f, e.cap));
14            if (d > 0) {
15                e.cap -= d, G[e.to][e.rev].cap += d;
16                return d;
17            }
18        }
19    }
20    return 0;
21 }
22 int ford_fulkerson(int s, int t) {
23     int flow = 0, f;
24     // for (int i = 0; i < n; i++) {
25     //     cout << i << " : ";
26     //     for (edge e: G[i])
27     //         cout << '(' << e.to << ', ' << e.cap << ')' <<
28     //         ' ';
29     //     cout << '\n';
30     // }
31     do {
32         memset(vis, false, sizeof(vis));
33         f = dfs(s, t, INF);
34         // for (int i = 0; i < n; i++) {
35         //     cout << i << " : ";
36         //     for (edge e: G[i])
37         //         cout << '(' << e.to << ', ' << e.cap << ')' <<
38         //         << ' ';
39         //     cout << '\n';
40         // }
41         // cout << f << '\n';
42         flow += f;
43     } while (f > 0);
44     return flow;
45 }
46 void init(int n) {
47     for (int i = 0; i < n; i++) G[i].clear();
48 }
49 int main() {
50     cin >> n >> m >> s >> t;
51     init(n);
52     while (m--) {
53         cin >> a >> b >> c;
54         G[a].push_back((edge){b, c, (int)G[b].size()});
55         G[b].push_back((edge){a, 0, (int)G[a].size() - 1});
56     }
57     cout << ford_fulkerson(s, t) << '\n';
58     return 0;
59 }

```

6.9 HLD [f57ec6]

```

1 #include <bits/stdc++.h>
2 #define int long long
3 using namespace std;
4
5 const int N = 100005;
6 vector<int> G[N];
7 struct HLD {
8     vector<int> pa, sz, depth, mxson, topf, id;

```

```

9     int n, idcnt = 0;
10    HLD(int _n) : n(_n), pa(_n + 1), sz(_n + 1), depth(_n +
11        1), mxson(_n + 1), topf(_n + 1), id(_n + 1) {}
12    void dfs1(int v = 1, int p = -1) {
13        pa[v] = p; sz[v] = 1; mxson[v] = 0;
14        depth[v] = (p == -1 ? 0 : depth[p] + 1);
15        for (int u : G[v]) {
16            if (u == p) continue;
17            dfs1(u, v);
18            sz[v] += sz[u];
19            if (sz[u] > sz[mxson[v]]) mxson[v] = u;
20        }
21    }
22    void dfs2(int v = 1, int top = 1) {
23        id[v] = ++idcnt;
24        topf[v] = top;
25        if (mxson[v]) dfs2(mxson[v], top);
26        for (int u : G[v]) {
27            if (u == mxson[v] || u == pa[v]) continue;
28            dfs2(u, u);
29        }
30    }
31    // query 為區間資料結構
32    int path_query(int a, int b) {
33        int res = 0;
34        while (topf[a] != topf[b]) { /// 若不在同一條鍊上
35            if (depth[topf[a]] < depth[topf[b]]) swap(a, b);
36            res = max(res, 0ll); // query : l = id[topf[a]],
37            r = id[a]
38            a = pa[topf[a]];
39        }
40        /// 此時已在同一條鍊上
41        if (depth[a] < depth[b]) swap(a, b);
42        res = max(res, 0ll); // query : l = id[b], r = id[a]
43        return res;
44    }
45 }

```

6.10 Kosaraju SCC [c7d5aa]

```

1 // 給定一個有向圖、迴傳傳縮點後的圖、SCC 的資訊
2 // 所有點都以 based-0 編號
3 // 函式：
4 // .compress: O(n Log n) 計算 G3、SCC、SCC_id 的資訊，並把縮
5 // 點後的結果存在 result 裡
6 // SCC[i] = 某個 SCC 中的所有點
7 // SCC_id[i] = 第 i 個點在第幾個 SCC
8 struct SCC_compress {
9     int N, M, sz;
10    vector<vector<int>> G, inv_G, result;
11    vector<pair<int, int>> edges;
12    vector<bool> vis;
13    vector<int> order;
14
15    vector<vector<int>> SCC;
16    vector<int> SCC_id;
17
18    SCC_compress(int _N) :
19        N(_N), M(0), sz(0),
20        G(N), inv_G(N),
21        vis(N), SCC_id(N)
22    {}
23    vector<int> operator [] (int x){

```

```

24     return result[x];
25 }
26
27 void add_edge(int u, int v){
28     G[u].push_back(v);
29     inv_G[v].push_back(u);
30     edges.push_back({u, v});
31     M++;
32 }
33
34 void dfs1(vector<vector<int>> &G, int now){
35     vis[now] = 1;
36     for (auto x : G[now]) if (!vis[x]) dfs1(G, x);
37     order.push_back(now);
38 }
39
40 void dfs2(vector<vector<int>> &G, int now){
41     SCC_id[now] = SCC.size() - 1;
42     SCC.back().push_back(now);
43     vis[now] = 1;
44     for (auto x : G[now]) if (!vis[x]) dfs2(G, x);
45 }
46
47 void compress(){
48     fill(vis.begin(), vis.end(), 0);
49     for (int i=0; i<N; i++) if (!vis[i]) dfs1(G, i);
50
51     fill(vis.begin(), vis.end(), 0);
52     reverse(order.begin(), order.end());
53     for (int i=0; i<N; i++){
54         if (!vis[order[i]]){
55             SCC.push_back(vector<int>());
56             dfs2(inv_G, order[i]);
57         }
58     }
59
60     result.resize(SCC.size());
61     sz = SCC.size();
62     for (auto [u, v] : edges){
63         if (SCC_id[u] != SCC_id[v]) result[SCC_id[u]].
64             push_back(SCC_id[v]);
65     }
66     for (int i=0; i<SCC.size(); i++){
67         sort(result[i].begin(), result[i].end());
68         result[i].resize(unique(result[i].begin(), result
69             [i].end())-result[i].begin());
70     }
71 }

```

6.11 Kuhn Munkres [e66c35]

```

1 // O(n^3) 找到最大權匹配
2 struct KuhnMunkres {
3     int n; // max(n, m)
4     vector<vector<int>> G;
5     vector<int> match, lx, ly, visx, visy;
6     vector<int> slack;
7     int stamp = 0;
8
9     KuhnMunkres(int n) : n(n), G(n, vector<int>(n)), lx(n),
10        ly(n), slack(n), match(n), visx(n), visy(n) {}
11
12     void add(int x, int y, int w){
13         G[x][y] = max(G[x][y], w);

```

```

13 }
14
15 bool dfs(int i, bool aug){ // aug = true 表示要更新 match
16     if (visx[i]==stamp) return false;
17     visx[i] = stamp;
18
19     for (int j=0 ; j<n ; j++){
20         if (visy[j]==stamp) continue;
21         int d = lx[i]+ly[j]-G[i][j];
22
23         if (d==0){
24             visy[j] = stamp;
25             if (match[j]==-1 || dfs(match[j], aug)){
26                 if (aug){
27                     match[j] = i;
28                 }
29                 return true;
30             }
31         }else{
32             slack[j] = min(slack[j], d);
33         }
34     }
35     return false;
36 }
37
38 bool augment(){
39     for (int j=0 ; j<n ; j++){
40         if (visy[j]!=stamp && slack[j]==0){
41             visy[j] = stamp;
42             if (match[j]==-1 || dfs(match[j], false)){
43                 return true;
44             }
45         }
46     }
47     return false;
48 }
49
50 void relabel(){
51     int delta = INF;
52     for (int j=0 ; j<n ; j++){
53         if (visy[j]!=stamp) delta = min(delta, slack[j]);
54     }
55     for (int i=0 ; i<n ; i++){
56         if (visx[i]==stamp) lx[i] -= delta;
57     }
58     for (int j=0 ; j<n ; j++){
59         if (visy[j]==stamp) ly[j] += delta;
60         else slack[j] -= delta;
61     }
62 }
63
64 int solve(){
65
66     for (int i=0 ; i<n ; i++){
67         lx[i] = 0;
68         for (int j=0 ; j<n ; j++){
69             lx[i] = max(lx[i], G[i][j]);
70         }
71     }
72
73     fill(ly.begin(), ly.end(), 0);
74     fill(match.begin(), match.end(), -1);
75
76     for (int i = 0; i < n; i++) {
77         fill(slack.begin(), slack.end(), INF);

```

```

78     stamp++;
79     if(dfs(i, true)) continue;
80
81     while(augment()==false) relabel();
82     stamp++;
83     dfs(i, true);
84 }
85
86 int ans = 0;
87 for (int j=0 ; j<n ; j++){
88     if (match[j]!=-1){
89         ans += G[match[j]][j];
90     }
91 }
92 return ans;
93 }
94 };

```

6.12 LCA [5b6a5b]

```

1 // 1-based · 可以支援森林 · 0 是超級源點 · 所有樹都要跟他建邊
2 struct Tree{
3     int N, M = 0, H;
4     vector<int> parent, dep;
5     vector<vector<int>> G, LCA;
6
7     Tree(int _N) : N(_N+1), H(__lg(N)+1), parent(N, -1), dep(N, 0),
8         LCA.resize(H, vector<int>(N, 0));
9 }
10
11 void add_edge(int u, int v){
12     M++;
13     G[u].push_back(v);
14     G[v].push_back(u);
15 }
16
17 void dfs(int now = 0, int pre = 0){
18     dep[now] = dep[pre]+1;
19     parent[now] = pre;
20     for (auto x : G[now]){
21         if (x==pre) continue;
22         dfs(x, now);
23     }
24 }
25
26 void build_LCA(int root = 0){
27     dfs();
28     for (int i=0 ; i<N ; i++) LCA[0][i] = parent[i];
29     for (int i=1 ; i<H ; i++){
30         for (int j=0 ; j<N ; j++){
31             LCA[i][j] = LCA[i-1][LCA[i-1][j]];
32         }
33     }
34 }
35
36 int jump(int u, int step){
37     for (int i=0 ; i<H ; i++){
38         if (step&(1<<i)) u = LCA[i][u];
39     }
40     return u;
41 }
42
43 int get_LCA(int u, int v){
44     if (dep[u]<dep[v]) swap(u, v);

```

```

45     u = jump(u, dep[u]-dep[v]);
46     if (u==v) return u;
47     for (int i=H-1 ; i>=0 ; i--){
48         if (LCA[i][u]!=LCA[i][v]){
49             u = LCA[i][u];
50             v = LCA[i][v];
51         }
52     }
53     return parent[u];
54 }
55 };

```

6.13 MCMF [0d5244]

```

1 struct Flow {
2     struct Edge {
3         int u, rc, k, rv;
4     };
5
6     vector<vector<Edge>> G;
7     vector<int> par, par_eid;
8     Flow(int n) : G(n+1), par(n+1), par_eid(n+1) {}
9
10    // v->u, capacity: c, cost: k
11    void add(int v, int u, int c, int k){
12        G[v].push_back({u, c, k, G[u].size()});
13        G[u].push_back({v, 0, -k, G[v].size()-1});
14    }
15
16    // 6d1140
17    int spfa(int s, int t){
18        fill(par.begin(), par.end(), -1);
19        vector<int> dis(par.size(), INF);
20        vector<bool> in_q(par.size(), false);
21        queue<int> Q;
22        dis[s] = 0;
23        in_q[s] = true;
24        Q.push(s);
25
26        while (!Q.empty()){
27            int v = Q.front();
28            Q.pop();
29            in_q[v] = false;
30
31            for (int i=0 ; i<G[v].size() ; i++){
32                auto [u, rc, k, rv] = G[v][i];
33                if (rc>0 && dis[v]+k<dis[u]){
34                    dis[u] = dis[v]+k;
35                    par[u] = v;
36                    par_eid[u] = i;
37                    if (!in_q[u]) Q.push(u);
38                    in_q[u] = true;
39                }
40            }
41        }
42
43        return dis[t];
44    }
45
46    // return <max flow, min cost>, d7e7ad
47    pair<int, int> flow(int s, int t){
48        int fl = 0, cost = 0, d;
49        while ((d = spfa(s, t))<INF){
50            int cur = INF;
51            for (int v=t ; v!=s ; v=par[v])

```

```

52     cur = min(cur, G[par[v]][par_eid[v]].rc);
53     fl += cur;
54     cost += d*cur;
55     for (int v=t ; v!=s ; v=par[v]){
56         G[par[v]][par_eid[v]].rc -= cur;
57         G[v][G[par[v]][par_eid[v]].rv].rc += cur;
58     }
59 }
60 return {fl, cost};
61 }
62 };

```

6.14 Tarjan SCC [8b2350]

```

1 struct tarjan_SCC {
2     int now_T, now_SCCs;
3     vector<int> dfn, low, SCC;
4     stack<int> S;
5     vector<vector<int>> E;
6     vector<bool> vis, in_stack;
7
8     tarjan_SCC(int n) {
9         init(n);
10    }
11    void init(int n) {
12        now_T = now_SCCs = 0;
13        dfn = low = SCC = vector<int>(n);
14        E = vector<vector<int>>(n);
15        S = stack<int>();
16        vis = in_stack = vector<bool>(n);
17    }
18    void add(int u, int v) {
19        E[u].push_back(v);
20    }
21    void build() {
22        for (int i = 0; i < dfn.size(); ++i) {
23            if (!dfn[i]) dfs(i);
24        }
25    }
26    void dfs(int v) {
27        now_T++;
28        vis[v] = in_stack[v] = true;
29        dfn[v] = low[v] = now_T;
30        S.push(v);
31        for (auto &i:E[v]) {
32            if (!vis[i]) {
33                vis[i] = true;
34                dfs(i);
35                low[v] = min(low[v], low[i]);
36            }
37            else if (in_stack[i]) {
38                low[v] = min(low[v], dfn[i]);
39            }
40        }
41        if (low[v] == dfn[v]) {
42            int tmp;
43            do {
44                tmp = S.top();
45                S.pop();
46                SCC[tmp] = now_SCCs;
47                in_stack[tmp] = false;
48            } while (tmp != v);
49            now_SCCs += 1;
50        }
51    }

```

```
52| };
```

6.15 Theorem

- 任意圖
 - 最大匹配 + 最小邊覆蓋 = n (不能有孤點)
 - 點覆蓋的補集是獨立集。最小點覆蓋 + 最大獨立集 = n
 - w (最小權點覆蓋) + w (最大權獨立集) = $\sum w_v$
 - (帶點權的二分圖可以用最小割解。構造請參考 Augment Path.cpp)

- 二分圖

- 最小點覆蓋 = 最大匹配 = n - 最大獨立集

- 只有邊帶權的二分圖

- w-vertex-cover (帶權點覆蓋): 每條邊的兩個連接點被選中的次數總和至少要是 w_e 。
- w-weight matching (帶權匹配)
- minimum vertex count of w-vertex-cover = maximum weight count of w-weight matching (一個點可以被選很多次。但邊不行)

- 點、邊帶權的二分圖的定理

- b-matching: 假設 v 的點權是 b_v 。那所有 v 的匹配邊 e 的權重都要滿足 $\sum w_e \leq b_v$ 。
- The maximum w-weight of a b-matching equals the minimum b-weight of vertices in a w-vertex-cover.

6.16 Tree Isomorphism [6f5c9e]

```

1 // Please change "mul" to a random number (> N) before submit
2 struct Tree_Hash { // 1-based
3     using T = __int128;
4     const T mod = (T)(1e18) + 3, mul = 435216;
5     vector<vector<int>> E;
6     Tree_Hash(int _n) : E(_n + 1) {}
7     void add_edge(int u, int v) {
8         E[u].push_back(v); E[v].push_back(u);
9     }
10    pair<T, T> h(int v, int p = -1) {
11        vector<pair<T, T>> s;
12        for (auto &i:E[v]) if (i != p) {
13            s.push_back(h(i, v));
14        }
15        T res = s.size() + 1, prod = mul;
16        sort(s.begin(), s.end());
17        for (auto &[i, j] : s) {
18            res = (res + i * prod) % mod;
19            prod = prod * j % mod;
20        }
21        return {res, prod};
22    }
23    T cal(int v = 1) {
24        int n = 0, c1 = -1, c2 = -1;
25        auto dfs = [&](auto&& f, int v, int p) -> int {
26            int sz = 1, mx = 0;
27            for (int u : E[v]) if (u != p) {
28                int s = f(f, u, v);
29                sz += s;
30                mx = max(mx, s);
31            }

```

```

32         mx = max(mx, n - sz);
33         if (n && mx * 2 <= n) (~c1 ? c2 : c1) = v;
34         return sz;
35     };
36     n = dfs(dfs, v, -1); dfs(dfs, v, -1);
37     E.push_back({c1, (c2 == -1) ? c1 : c2});
38     return h(E.size() - 1).first;
39 }
40 };

```

6.17 prufer reverse [2f831d]

```

1 // K_n 有  $n^{n-2}$  種生成樹
2 vector<pair<int, int>> prufer_reverse(vector<int> &v) {
3     int n = v.size() + 2, mn = 1, mx = n; // 1-based
4     vector<int> deg(mx + 1);
5     for (auto &i:v) deg[i] += 1;
6     int ind = mn;
7     while (deg[ind] != 0) ind += 1;
8     int leaf = ind;
9     vector<pair<int, int>> res;
10    for (auto &i:v) {
11        res.emplace_back(leaf, i);
12        deg[i] -= 1;
13        if (deg[i] == 0 && i < ind) leaf = i;
14        else {
15            do ind += 1; while (deg[ind] != 0);
16            leaf = ind;
17        }
18    }
19    res.emplace_back(leaf, mx);
20    return res;
21 }

```

6.18 圖方樹 [5a6c9f]

```

1 struct BCC_AP { // 0-based, remember to build, 不支援重邊
2     int n, nbcc; // 注意孤點不會有任何方點連接
3     vector<vector<int>> E, F; // id >= n: 方點
4     vector<int> pa, dep, low, stk, paf, depf;
5     void dfs(int v, int p) {
6         dep[v] = low[v] = ~p ? dep[p] + 1 : 0;
7         stk.push_back(v), pa[v] = p;
8         for (auto& u : E[v]) if (u != p) {
9             if (low[u] == -1) {
10                dfs(u, v), low[v] = min(low[v], low[u]);
11                if (low[u] >= dep[v]) {
12                    int id = nbcc++, x;
13                    do {
14                        x = stk.back(), stk.pop_back();
15                        F[id + n].push_back(x), F[x].push_back(id + n);
16                    } while (x != u);
17                    F[id + n].push_back(v), F[v].push_back(id + n);
18                }
19            } else low[v] = min(low[v], dep[u]);
20        }
21    }
22    bool is_bridge(int u, int v) {
23        return (F[bcc_id(u, v) + n].size() == 2);
24    }
25    // 判斷原圖上的一個點是不是割點
26    bool is_cut(int x) { return F[x].size() != 1; }
    // 回傳第 id 個 bcc 的每個點 (0-based)

```

```

27 | vector<int> bcc(int id) { return F[id + n]; }
28 | // 對於邊 (u, v) 回傳包含它的 bcc id (每條邊都恰好被一個
    | bcc 包含)
29 | int bcc_id(int u, int v) { // starts from 0
30 |     return paf[depf[u] < depf[v] ? v : u] - n; }
31 | // 計算在新圖上的 depf, paf 陣列・呼叫 bcc_id() 的時候會
    | 用到
32 | void dfs2(int v, int p) {
33 |     depf[v] = ~p ? depf[p] + 1 : 0, paf[v] = p;
34 |     for (int u : F[v]) if (u != p) dfs2(u, v);
35 | }
36 | void build() {
37 |     low.assign(n, -1);
38 |     for (int i = 0; i < n; ++i) if (low[i] == -1)
39 |         dfs(i, -1), dfs2(i, -1);
40 | }
41 | void add_edge(int u, int v) {
42 |     E[u].push_back(v), E[v].push_back(u);
43 | }
44 | BCC_AP (int _n) : n(_n), nbcc(0), E(n), F(2 * n), pa(n),
    |     dep(n), low(n), stk(), paf(n * 2), depf(n * 2) {}
45 | };

```

6.19 最大權閉合圖 [6ca663]

```

1 | // 邊 u → v 表示選 u 就要選 v (0-based)
2 | // 保證回傳值非負
3 | // 構造：呼叫 construct・選那些 vis 是 true 的點・
4 | // 一般圖：O(n²m) / 二分圖：O(m√n)
5 | template<typename U>
6 | U maximum_closure(vector<U> w, vector<pair<int,int>> EV) {
7 |     int n = w.size(), S = n + 1, T = n + 2;
8 |     Flow G(T + 5); // Graph/Dinic.cpp
9 |     U sum = 0;
10 |    for (int i = 0; i < n; ++i) {
11 |        if (w[i] > 0) {
12 |            G.add(S, i, w[i]);
13 |            sum += w[i];
14 |        }
15 |        else if (w[i] < 0) {
16 |            G.add(i, T, abs(w[i]));
17 |        }
18 |    }
19 |    for (auto &[u, v] : EV) { // 請務必確保 INF > Σ|w_i|
20 |        G.add(u, v, INF);
21 |    }
22 |    U cut = G.flow(S, T);
23 |    return sum - cut;
24 | }

```

7 Math

7.1 Burnside's Lemma

$$\sum_{k=1}^n \frac{c(k)}{n}$$

- n : 有多少種置換方式 (例如：旋轉方式)
- $c(k)$: 所有可能中・經過 k 次旋轉後・仍不會和別人相同的方式的數量

7.2 CRT [6a6da6]

```

1 | using ll = __int128;
2 | // a * p.first + b * p.second = gcd(a, b)
3 | pair<ll, ll> extgcd(ll a, ll b) {
4 |     if (b == 0) return {1, 0};
5 |     auto [y, x] = extgcd(b, a % b);
6 |     return pair<ll, ll>(x, y - (a / b) * x);
7 | }
8 |
9 | pair<ll, ll> CRT(ll x1, ll m1, ll x2, ll m2) {
10 |    ll g = __gcd(m1, m2);
11 |    if ((x2 - x1) % g) return make_pair(-1, -1); // no sol
12 |    m1 /= g, m2 /= g;
13 |    pair<ll, ll> p = extgcd(m1, m2);
14 |    ll lcm = m1 * m2 * g;
15 |    ll res = p.first * (x2 - x1) * m1 + x1;
16 |    // be careful with overflow
17 |    return make_pair((res % lcm + lcm) % lcm, lcm);
18 | }
19 |
20 | pair<ll, ll> CRT_arr(vector<ll> &a, vector<ll> &m) {
21 |    pair<ll, ll> res(a[0], m[0]);
22 |    for (int i = 1; i < a.size(); ++i) {
23 |        res = CRT(res.first, res.second, a[i], m[i]);
24 |        if (res.first == -1) break;
25 |    }
26 |    return res;
27 | }

```

7.3 Catalan Number

任意括號序列： $C_n = \frac{1}{n+1} \binom{2n}{n}$

7.4 Floor Sum [5d1c9e]

```

1 | // Returns sum_{i=0}^{n-1} floor((a*i + b) / m)
2 | // Log( max(a, m) )
3 | int floor_sum(int n, int m, int a, int b) {
4 |     int ans = 0;
5 |     while (true) {
6 |         if (a >= m) {
7 |             ans += (n - 1) * n * (a / m) / 2;
8 |             a %= m;
9 |         }
10 |        if (b >= m) {
11 |            ans += n * (b / m);
12 |            b %= m;
13 |        }
14 |        int y_max = a * n + b;
15 |        if (y_max < m) break;
16 |        n = y_max / m;
17 |        b = y_max % m;
18 |        swap(m, a);
19 |    }
20 |    return ans;
21 | }

```

7.5 Josephus Problem [e0ed50]

```

1 | // 有 n 個人・第偶數個報數的人被刪掉・問第 k 個被踢掉的是誰
2 | int solve(int n, int k){
3 |     if (n==1) return 1;
4 |     if (k<=(n+1)/2){
5 |         if (2*k>n) return 2*k%n;

```

```

6 |         else return 2*k;
7 |     }else{
8 |         int res=solve(n/2, k-(n+1)/2);
9 |         if (n&1) return 2*res+1;
10 |        else return 2*res-1;
11 |     }
12 | }

```

7.6 Lagrange any x [1f2c26]

```

1 | // init: (x1, y1), (x2, y2) in a vector
2 | struct Lagrange{
3 |     int n;
4 |     vector<pair<int, int>> v;
5 |
6 |     Lagrange(vector<pair<int, int>> &_v){
7 |         n = _v.size();
8 |         v = _v;
9 |     }
10 |
11 |    // O(n^2 Log MAX_A)
12 |    int solve(int x){
13 |        int ret = 0;
14 |        for (int i=0; i<n; i++){
15 |            int now = v[i].second;
16 |            for (int j=0; j<n; j++){
17 |                if (i==j) continue;
18 |                now *= ((x-v[j].first+MOD)%MOD);
19 |                now %= MOD;
20 |                now *= (qp((v[i].first-v[j].first+MOD)%MOD,
21 |                    MOD-2)+MOD)%MOD;
22 |                now %= MOD;
23 |            }
24 |            ret = (ret+now)%MOD;
25 |        }
26 |        return ret;
27 |    }
28 | };

```

7.7 Lagrange continuous x [57536a]

```

1 | #include <bits/stdc++.h>
2 | using namespace std;
3 |
4 | const int MAX_N = 5e5 + 10;
5 | const int mod = 1e9 + 7;
6 |
7 | long long inv_fac[MAX_N];
8 |
9 | inline int fp(long long x, int y) {
10 |    int ret = 1;
11 |    for (; y; y >= 1) {
12 |        ret = (y & 1) ? (ret * x % mod) : ret;
13 |        x = x * x % mod;
14 |    }
15 |    return ret;
16 | }
17 |
18 | // TO USE THIS TEMPLATE, YOU MUST MAKE SURE THAT THE MOD
    | NUMBER IS A PRIME.
19 | struct Lagrange {
20 |     /*
21 |        Initialize a polynomial with f(x_0), f(x_0 + 1), ..., f(
    | x_0 + n).

```

```

22 This determines a polynomial f(x) whose degree is at most 83 }
23 n.
24 Then you can call sample(x) and you get the value of f(x)
25 .
26 Complexity of init() and sample() are both O(n).
27 */
28 int m, shift; // m = n + 1
29 vector<int> v, mul;
30 // You can use this function if you don't have inv_fac array
31 already.
32 void construct_inv_fac() {
33     long long fac = 1;
34     for (int i = 2; i < MAX_N; ++i) {
35         fac = fac * i % mod;
36     }
37     inv_fac[MAX_N - 1] = fp(fac, mod - 2);
38     for (int i = MAX_N - 1; i >= 1; --i) {
39         inv_fac[i - 1] = inv_fac[i] * i % mod;
40     }
41 // You call init() many times without having a second
42 instance of this struct.
43 void init(int X_0, vector<int> &u) {
44     v = u;
45     shift = ((1 - X_0) % mod + mod) % mod;
46     if (v.size() == 1) v.push_back(v[0]);
47     m = v.size();
48     mul.resize(m);
49 }
50 // You can use sample(x) instead of sample(x % mod).
51 int sample(int x) {
52     x = ((long long)x + shift) % mod;
53     x = (x < 0) ? (x + mod) : x;
54     long long now = 1;
55     for (int i = m; i >= 1; --i) {
56         mul[i - 1] = now;
57         now = now * (x - i) % mod;
58     }
59     int ret = 0;
60     bool neg = (m - 1) & 1;
61     now = 1;
62     for (int i = 1; i <= m; ++i) {
63         int up = now * mul[i - 1] % mod;
64         int down = inv_fac[m - i] * inv_fac[i - 1] % mod;
65         int tmp = ((long long)v[i - 1] * up % mod) * down
66             % mod;
67         ret += (neg && tmp) ? (mod - tmp) : (tmp);
68         ret = (ret >= mod) ? (ret - mod) : ret;
69         now = now * (x - i) % mod;
70         neg ^= 1;
71     }
72     return ret;
73 }
74 };
75
76 int main() {
77     int n; cin >> n;
78     vector<int> v(n);
79     for (int i = 0; i < n; ++i) {
80         cin >> v[i];
81     }
82     Lagrange L;
83     L.construct_inv_fac();
84     L.init(0, v);
85     int x; cin >> x;
86     cout << L.sample(x);

```

7.8 Linear Mod Inverse [ecf71e]

```

1 // 線性求 1-based a[i] 對 p 的乘法反元素
2 vector<int> s(n+1, 1), invS(n+1), invA(n+1);
3 for (int i=1; i<=n; i++) s[i] = s[i-1]*a[i]%p;
4 invS[n] = qp(s[n], p-2, p);
5 for (int i=n; i>=1; i--) invS[i-1] = invS[i]*a[i]%p;
6 for (int i=1; i<=n; i++) invA[i] = invS[i]*s[i-1]%p;

```

7.9 Lucas's Theorem [b37dcf]

```

1 // 對於很大的 C^n_{m} 對質數 p 取模。只要 p 不大就可以用。
2 int Lucas(int n, int m, int p){
3     if (m==0) return 1;
4     return (C(n%p, m%p, p)*Lucas(n/p, m/p, p)%p);
5 }

```

7.10 Matrix [8d1a23]

```

1 struct Matrix{
2     int n, m;
3     vector<vector<int>> arr;
4
5     Matrix(int _n, int _m){
6         n = _n;
7         m = _m;
8         arr.assign(n, vector<int>(m));
9     }
10
11     vector<int> & operator [] (int i){
12         return arr[i];
13     }
14
15     Matrix operator * (Matrix b){
16         Matrix ret(n, b.m);
17         for (int i=0; i<n; i++){
18             for (int j=0; j<b.m; j++){
19                 for (int k=0; k<m; k++){
20                     ret.arr[i][j] += arr[i][k]*b.arr[k][j]%
21                         MOD;
22                     ret.arr[i][j] %= MOD;
23                 }
24             }
25         }
26         return ret;
27     }
28
29     Matrix pow(int p){
30         Matrix ret(n, n), mul = *this;
31         for (int i=0; i<n; i++){
32             ret.arr[i][i] = 1;
33         }
34
35         for (; p; p>>=1){
36             if (p&1) ret = ret*mul;
37             mul = mul*mul;
38         }
39
40         return ret;
41     }
42 }

```

```

43 int det(){
44     vector<vector<int>> arr = this->arr;
45     bool flag = false;
46     for (int i=0; i<n; i++){
47         int target = -1;
48         for (int j=i; j<n; j++){
49             if (arr[j][i]){
50                 target = j;
51                 break;
52             }
53         }
54         if (target==-1) return 0;
55         if (i!=target){
56             swap(arr[i], arr[target]);
57             flag = !flag;
58         }
59
60         for (int j=i+1; j<n; j++){
61             if (!arr[j][i]) continue;
62             int freq = arr[j][i]*qp(arr[i][i], MOD-2)%MOD;
63             for (int k=i; k<n; k++){
64                 arr[j][k] -= freq*arr[i][k];
65                 arr[j][k] = (arr[j][k]%MOD+MOD)%MOD;
66             }
67         }
68     }
69
70     int ret = !flag ? 1 : MOD-1;
71     for (int i=0; i<n; i++){
72         ret *= arr[i][i];
73         ret %= MOD;
74     }
75     return ret;
76 }
77 };

```

7.11 Matrix 01 [8d542a]

```

1 const int MAX_N = (1LL<<12);
2 struct Matrix{
3     int n, m;
4     vector<bitset<MAX_N>> arr;
5
6     Matrix(int _n, int _m){
7         n = _n;
8         m = _m;
9         arr.resize(n);
10    }
11
12    Matrix operator * (Matrix b){
13        Matrix b_t(b.m, b.n);
14        for (int i=0; i<b.n; i++){
15            for (int j=0; j<b.m; j++){
16                b_t.arr[j][i] = b.arr[i][j];
17            }
18        }
19
20        Matrix ret(n, b.m);
21        for (int i=0; i<n; i++){
22            for (int j=0; j<b.m; j++){
23                ret.arr[i][j] = ((arr[i]&b_t.arr[j]).count()
24                    &1);
25            }
26        }
27    }

```

```

26         return ret;
27     }
28 };

```

7.12 Matrix Tree Theorem

目標：給定一張無向圖，問他的生成樹數量。
方法：先把所有自環刪掉，定義 Q 為以下矩陣

$$Q_{i,j} = \begin{cases} \deg(v_i) & \text{if } i = j \\ -(邊v_i v_j \text{ 的數量}) & \text{otherwise} \end{cases}$$

接著刪掉 Q 的第一個 row 跟 column，它的 determinant 就是答案。
目標：給定一張有向圖，問他的以 r 為根，可以走到所有點生成樹數量。

方法：先把所有自環刪掉，定義 Q 為以下矩陣

$$Q_{i,j} = \begin{cases} \deg_{in}(v_i) & \text{if } i = j \\ -(邊v_i v_j \text{ 的數量}) & \text{otherwise} \end{cases}$$

接著刪掉 Q 的第 r 個 row 跟 column，它的 determinant 就是答案。

7.13 Miller Rabin [5694bf]

```

1 // O(k log^3 n), k = llsprp.size()
2 using Uint = unsigned long long;
3 Uint modmul(Uint a, Uint b, Uint m) {
4     int ret = a*b - m*(Uint)((long double)a*b/m);
5     return ret + m*(ret < 0) - m*(ret >=(int)m);
6 }
7
8 int qp(int b, int p, int m){
9     int ret = 1;
10    for (; p ; p>>=1){
11        if (p&1) ret = modmul(ret, b, m);
12        b = modmul(b, b, m);
13    }
14    return ret;
15 }
16
17 // ed23aa
18 vector<int> llsprp = {2, 325, 9375, 28178, 450775, 9780504,
19                     1795265022};
20 bool is_prime(int n, vector<int> sprp = llsprp){
21     if (n==2) return 1;
22     if (n<2 || n%2==0) return 0;
23
24     int t = 0;
25     int u = n-1;
26     for (; u%2==0 ; t++) u>>=1;
27
28     for (int i=0 ; i<sprp.size() ; i++){
29         int a = sprp[i]%n;
30         if (a==0 || a==1 || a==n-1) continue;
31         int x = qp(a, u, n);
32         if (x==1 || x==n-1) continue;
33         for (int j=0 ; j<t ; j++){
34             x = modmul(x, x, n);
35             if (x==1) return 0;
36             if (x==n-1) break;
37         }
38         if (x==n-1) continue;

```

```

39         return false;
40     }
41
42     return true;
43 }

```

7.14 Pollard Rho [a5daef]

```

1 mt19937 seed(chrono::steady_clock::now().time_since_epoch().
2               count());
3 int rnd(int l, int r){
4     return uniform_int_distribution<int>(l, r)(seed);
5 }
6 // O(n^{1/4}) 回傳 1 或自己的因數、記得先判斷 n 是不是質數
7 // (用 Miller-Rabin)
8 // c1670c
9 int Pollard_Rho(int n){
10    int s = 0, t = 0;
11    int c = rnd(1, n-1);
12
13    int step = 0, goal = 1;
14    int val = 1;
15
16    for (goal=1 ; ; goal<=1, s=t, val=1){
17        for (step=1 ; step<=goal ; step++){
18            t = ((__int128)t*t+c)%n;
19            val = ((__int128)val*abs(t-s)%n);
20
21            if ((step % 127) == 0){
22                int d = __gcd(val, n);
23                if (d>1) return d;
24            }
25        }
26
27        int d = __gcd(val, n);
28        if (d>1) return d;
29    }
30 }

```

7.15 Stirling's formula

$$n! \approx \sqrt{2\pi n} \left(\frac{n}{e}\right)^n$$

7.16 Theorem

- $1 \sim x$ 質數的數量 $\approx \frac{x}{\ln x}$
- x 的因數的數量 $\approx x^{\frac{1}{3}}$
- x 的質因數的數量 $\approx \log \log x$
- p is a prime number $\Leftrightarrow (p-1)! \equiv -1 \pmod{p}$
- 每個正整數都可以表示成四個整數的平方和
- 任何大於 2 的整數都可以表示成兩個質數的和
- $n^{k-2} \cdot \prod_{i=1}^k s_i$ n 個點、 k 的連通塊，加上 $k-1$ 條邊使得變成一個連通圖的方法數，其中每個連通塊有 s_i 個點

7.17 josephus [0be067]

```

1 // n 個人，每 k 個人就刪除的約瑟夫遊戲
2 int josephus(int n, int k) {
3     if (n == 1)
4         return 0;
5     if (k == 1)
6         return n-1;
7     if (k > n)
8         return (josephus(n-1, k) + k) % n;
9     int cnt = n / k;
10    int res = josephus(n - cnt, k);
11    res -= n % k;
12    if (res < 0)
13        res += n;
14    else
15        res += res / (k - 1);
16    return res;
17 }

```

7.18 二元一次方程式

$$\begin{cases} ax + by = e \\ cx + dy = f \end{cases} = \begin{cases} x = \frac{ed-bf}{ad-bc} \\ y = \frac{af-ec}{ad-bc} \end{cases}$$

若 $x = \frac{0}{0}$ 且 $y = \frac{0}{0}$ ，則代表無限多組解。若 $x = \frac{*}{0}$ 且 $y = \frac{*}{0}$ ，則代表無解。

7.19 數論分塊 [8ccab5]

```

1 // 時間複雜度為 O(sqrt(n))
2 // 區間為 [L, r]
3 for(int i=1 ; i<=n ; i++){
4     int l = i, r = n/(n/i);
5     i = r;
6     ans.push_back(r);
7 }

```

7.20 最大質因數 [ca5e52]

```

1 void max_fac(int n, int &ret){
2     if (n<=ret || n<2) return;
3     if (isprime(n)){
4         ret = max(ret, n);
5         return;
6     }
7
8     int p = Pollard_Rho(n);
9     max_fac(p, ret), max_fac(n/p, ret);
10 }

```

7.21 歐拉公式 [7e25f3]

```

1 // phi(n) = 小於 n 並與 n 互質的正整數數量。
2 // O(sqrt(n))，回傳 phi(n)
3 int phi(int n){
4     int ret = n;
5
6     for (int i=2 ; i*i<=n ; i++){
7         if (n%i==0){
8             while (n%i==0) n /= i;
9             ret = ret*(i-1)/i;
10        }
11    }

```

```

12     if (n>1) ret = ret*(n-1)/n;
13
14     return ret;
15 }
16
17 // O(n) · 回傳 1~n 的 phi 值
18 vector<int> phi_1_to_n(int n) {
19     vector<int> primes, lpf(n + 1), phi(n + 1);
20     phi[0] = 0; phi[1] = 1;
21     for (int i = 2; i <= n; ++i) {
22         if (lpf[i] == 0) {
23             primes.push_back(i);
24             lpf[i] = i;
25         }
26         for (auto p : primes) {
27             if (i * p > n) break;
28             lpf[i * p] = p;
29             if (i % p == 0) break;
30         }
31         bool b = i % (lpf[i] * lpf[i]);
32         phi[i] = phi[i / lpf[i]] * (lpf[i] - b);
33     }
34     return phi;
35 }

```

7.22 歐拉定理

$$a^n \equiv \begin{cases} a^{n \bmod \varphi(m)}, & \gcd(a, m) = 1, \\ a^n, & \gcd(a, m) \neq 1, n < \varphi(m), \\ a^{(n \bmod \varphi(m)) + \varphi(m)}, & \gcd(a, m) \neq 1, n \geq \varphi(m). \end{cases} \pmod{m}$$

7.23 錯排公式

錯排公式 : (n 個人中 · 每個人皆不再原來位置的組合數)

$$dp_i = \begin{cases} 1 & i = 0 \\ 0 & i = 1 \\ (i-1)(dp_{i-1} + dp_{i-2}) & \text{otherwise} \end{cases}$$

8 String

8.1 AC automation [018290]

```

1 struct ACAutomation{
2     vector<vector<int>> go;
3     vector<int> fail, match, pos;
4     int sz = 0; // 有效節點為 [0, sz] · 開陣列的時候要小
5     // 心 !!!
6     ACAutomation(int n) : go(n, vector<int>(26)), fail(n),
7         match(n) {}
8
9     void add(string s){
10         int now = 0;
11         for (char c : s){
12             if (!go[now][c-'a']) go[now][c-'a'] = ++sz;
13             now = go[now][c-'a'];
14         }
15         pos.push_back(now);
16     }
17
18     void build(){
19         queue<int> que;

```

```

19         for (int i=0 ; i<26 ; i++){
20             if (go[0][i]) que.push(go[0][i]);
21         }
22         while (que.size()){
23             int u = que.front();
24             que.pop();
25             for (int i=0 ; i<26 ; i++){
26                 if (go[u][i]){
27                     fail[go[u][i]] = go[fail[u]][i];
28                     que.push(go[u][i]);
29                 }else go[u][i] = go[fail[u]][i];
30             }
31         }
32     }
33
34     // counting pattern
35     void buildMatch(string &s){
36         int now = 0;
37         for (char c : s){
38             now = go[now][c-'a'];
39             match[now]++;
40         }
41     }
42
43     vector<int> in(sz+1), que;
44     for (int i=1 ; i<=sz ; i++) in[fail[i]]++;
45     for (int i=1 ; i<=sz ; i++) if (in[i]==0) que.
46         push_back(i);
47     for (int i=0 ; i<que.size() ; i++){
48         int now = que[i];
49         match[fail[now]] += match[now];
50         if (--in[fail[now]]==0) que.push_back(fail[now]);
51     }
52 }

```

8.2 Enumerate Runs [74311c]

```

1 // returns [minimum_period, ans_l, ans_r] (閉區間, 0-based)
2 // (ans_r - ans_l + 1) >= minimum_period * 2
3 vector<array<int, 3>> enumerate_run(string s) {
4     int n = s.size();
5     if (n <= 1) return {};
6     vector<vector<pair<int, int>>> stor(n + 1);
7
8     auto sub = [&](string l, string r) {
9         int a = l.size(), b = r.size();
10        auto zl = z_function(string(l.rbegin(), l.rend()));
11        auto zr = z_function(r + l + r); // Z_Algorithm.cpp
12        vector<array<int, 3>> res;
13        for (int p = 1; p <= a; p++) {
14            int x = (p == a) ? p : min(zl[p] + p, a);
15            int y = min(zr[a + b - p], b);
16            if (x + y >= 2 * p) res.push_back({p, x, y});
17        }
18        return res;
19    };
20
21    auto dfs = [&](auto self, int l, int r) -> void {
22        if (r - l <= 1) return;
23        int m = (l + r) >> 1;
24        self(self, l, m); self(self, m, r);
25        auto sl = s.substr(l, m-l), sr = s.substr(m, r-m);
26        for (auto [p, a, b] : sub(sl, sr))
27            stor[p].push_back({m - a, m + b});
28        reverse(sl.begin(), sl.end());

```

```

29        reverse(sr.begin(), sr.end());
30        for (auto [p, a, b] : sub(sr, sl))
31            stor[p].push_back({m - b, m + a});
32    };
33    dfs(dfs, 0, n);
34
35    vector<array<int, 3>> runs;
36    set<pair<int, int>> done;
37    for (int p = 1; p <= n; p++) {
38        auto& v = stor[p];
39        sort(v.begin(), v.end(), [](auto& x, auto& y) {
40            return (x.first != y.first) ? (x < y)
41                : (x.second > y.second);
42        });
43        int mx = -1;
44        for (auto &[l, r] : v) if (r > mx) {
45            mx = r;
46            if (done.insert({l, r}).second)
47                runs.push_back({p, l, r - 1});
48        }
49    }
50    sort(runs.begin(), runs.end());
51    return runs;
52 }

```

8.3 Hash [942f42]

```

1 mt19937 seed(chrono::steady_clock::now().time_since_epoch().
2     count());
3 int rng(int l, int r){
4     return uniform_int_distribution<int>(l, r)(seed);
5 }
6 int A = rng(1e5, 8e8);
7 const int B = 1e9+7;
8
9 // 2f6192
10 struct RollingHash{
11     vector<int> Pow, Pre;
12     RollingHash(string s = ""){
13         Pow.resize(s.size());
14         Pre.resize(s.size());
15
16         for (int i=0 ; i<s.size() ; i++){
17             if (i==0){
18                 Pow[i] = 1;
19                 Pre[i] = s[i];
20             }else{
21                 Pow[i] = Pow[i-1]*A%B;
22                 Pre[i] = (Pre[i-1]*A+s[i])%B;
23             }
24         }
25         return;
26     }
27
28     int get(int l, int r){ // 取得 [l, r] 的數值
29         if (l==0) return Pre[r];
30         int res = (Pre[r]-Pre[l-1]*Pow[r-l+1])%B;
31         if (res<0) res += B;
32         return res;
33     }
34 };

```

8.4 KMP [287490]

```

1 // KMP[i] = s[0...i] 的最長共同前後綴長度 · KMP[KMP[i]-1] 可以
  跳 fail Link
2 vector<int> KMP(string &s){
3     vector<int> ret(s.size());
4     for (int i=1 ; i<s.size() ; i++){
5         int j = ret[i-1];
6         while (j && s[i]!=s[j]) j = ret[j-1];
7         ret[i] = j + (s[i]==s[j]);
8     }
9     return ret;
10 }

```

8.5 Manacher [199269]

```

1 // 回傳所有最長 (無法往兩側擴展) 迴文的區間 [L, r], 0-based
2 vector<array<int, 2>> Manacher(string str) {
3     string tmp = "$#";
4     for(char i : str) {
5         tmp += i;
6         tmp += '#';
7     }
8
9     int m = tmp.size(), mx = 0, id = 0;
10    vector<int> p(m);
11    vector<array<int, 2>> res;
12    for(int i=1 ; i<m ; i++) {
13        p[i] = mx > i ? min(p[id*2-i], mx-i) : 1;
14        while(tmp[i+p[i]] == tmp[i-p[i]]) p[i]++;
15        if (mx < i + p[i]) mx = i + p[i], id = i;
16
17        if (p[i] > 1) {
18            int r = (i / 2) + (p[i] / 2) - 2 + (i & 1);
19            res.push_back({r - p[i] + 2, r});
20        }
21    }
22    return res;
23 }

```

8.6 Min Rotation [b24786]

```

1 int minRotation(string s) {
2     int a = 0, n = s.size();
3     s += s;
4
5     for (int b=0 ; b<n ; b++){
6         for (int k=0 ; k<n ; k++){
7             if (a+k==b || s[a+k]<s[b+k]){
8                 b += max(0LL, k-1);
9                 break;
10            }
11            if (s[a+k]>s[b+k]){
12                a = b;
13                break;
14            }
15        }
16    }
17    return a;
18 }

```

8.7 Suffix Array [47a062]

```

1 // 注意 · 當 |s|=1 時 · lcp 不會有值 · 務必測試 |s|=1 的 case
2 struct SuffixArray {
3     string s;
4     vector<int> sa, lcp;
5
6     // 0eea96
7     template<typename T>
8     void sais(const T s, int n, int *sa, int *lar, int *p,
9               int *t, int A) {
10        int *rnk = p + n, *q = p + n / 2, *bkt = lar + A;
11        int m = 0, i, j, x = t[n - 1] = 1, y = rnk[0] = -1,
12            cnt = -1;
13
14        for (i=n-2 ; ~i ; i--) t[i] = (s[i] == s[i + 1] ? t[i
15            + 1] : s[i] < s[i + 1]);
16        for (i=1 ; i<n ; i++) rnk[i] = t[i] && !t[i-1] ? (p[m
17            ]=i, m++) : -1;
18        fill_n(lar, A, 0);
19        for (i=0 ; i<n ; i++) ++lar[s[i]];
20        for (i=1 ; i<A ; i++) lar[i] += lar[i-1];
21        auto pushS = [&](int x) { sa[--bkt[s[x]]] = x; };
22        auto pushL = [&](int x) { sa[bkt[s[x]]++] = x; };
23        auto induce_sort = [&](int* v) {
24            fill_n(sa, n, 0);
25            copy_n(lar, A, bkt);
26            for (i=m-1 ; ~i ; i--) pushS(v[i]);
27            copy_n(lar, A - 1, bkt + 1);
28            for (i=0 ; i<n ; i++) if (sa[i] && !t[sa[i] - 1])
29                pushL(sa[i] - 1);
30            copy_n(lar, A, bkt);
31            for (i=n-1 ; ~i ; i--) if (sa[i] && t[sa[i] - 1])
32                pushS(sa[i] - 1);
33        };
34        induce_sort(p);
35        for (i=0 ; i<n ; i++){
36            if (~(x=rnk[sa[i]])){
37                j = y < 0 || memcmp(s + p[x], s + p[y], (p[x
38                    + 1] - p[x]) * sizeof(s[0]));
39                q[y = x] = cnt += j;
40            }
41        }
42        if (cnt+1<m) sais(q, m, sa, bkt, rnk, t + n, cnt + 1)
43            ;
44        else for (i=0 ; i<m ; i++) sa[q[i]] = i;
45        for (i=0 ; i<m ; i++) q[i] = p[sa[i]];
46        induce_sort(q);
47    }
48
49    // da9ddf, O(n + Lim), Lim = max{s[i]}
50    template<typename T>
51    SuffixArray(const T& _s) : s(_s.begin(), _s.end()) {
52        s.push_back(-1);
53        for (auto &i:s) i += 1;
54        int n = s.size(), lim = *max_element(s.begin(), s.end
55            ()) + 5;
56        sa.resize(n);
57        lcp.resize(n);
58        vector<int> bkt(n + lim * 2), p(n * 2), t(n * 2),
59            rank(n);
60        sais(&s[0], n, &sa[0], &bkt[0], &p[0], &t[0], lim);
61
62        for (int i=1 ; i<n ; i++) rank[sa[i]] = i;
63        for (int i=0, j, k=0 ; i<n-1 ; lcp[rank[i++]]=k){
64

```

```

65            for (k && k--, j=sa[rank[i]-1] ; i+k<s.size() &&
66                j+k<s.size() && s[i+k]==s[j+k] ; k++);
67        }
68        sa.erase(sa.begin());
69        lcp.erase(lcp.begin(), lcp.begin()+2);
70        s.pop_back();
71    }
72
73    // f49583
74    vector<int> pos; // pos[i] = i 這個值在 pos 的哪個地方
75    SparseTable st;
76    void init_lcp(){
77        pos.resize(sa.size());
78        for (int i=0 ; i<sa.size() ; i++){
79            pos[sa[i]] = i;
80        }
81        if (lcp.size()){
82            st.build(lcp);
83        }
84    }
85
86    // 用之前記得 init
87    // 查詢「sa 上的位置」的 x 跟 y 的 lcp
88    int get_lcp(int x, int y){
89        if (x==y) return s.size()-x;
90        if (x>y) swap(x, y);
91        return st.query(x, y);
92    }
93
94    // 回傳 [l1, r1] 跟 [l2, r2] 的 lcp · 0-based
95    int get_lcp(int l1, int r1, int l2, int r2){
96        int pos_1 = pos[l1], len_1 = r1-l1+1;
97        int pos_2 = pos[l2], len_2 = r2-l2+1;
98        if (pos_1>pos_2){
99            swap(pos_1, pos_2);
100            swap(len_1, len_2);
101        }
102        if (l1==l2){
103            return min(len_1, len_2);
104        }else{
105            return min({st.query(pos_1, pos_2), len_1, len_2
106                });
107        }
108    }
109
110    // 檢查 [l1, r1] 跟 [l2, r2] 的大小關係 · 0-based
111    // 如果前者小於後者 · 就回傳 <0 · 相等就回傳 =0 · 否則回傳
112    >0
113    // 5b8db0
114    int substring_cmp(int l1, int r1, int l2, int r2){
115        int len_1 = r1-l1+1;
116        int len_2 = r2-l2+1;
117        int res = get_lcp(l1, r1, l2, r2);
118
119        if (res<len_1 && res<len_2){
120            return s[l1+res]-s[l2+res];
121        }else if (len_1==res && len_2==res){
122            return 0;
123        }else{
124            return len_1==res ? -1 : 1;
125        }
126    }

```

```

116 // 對於位置在 <=p 的後綴，找離他左邊/右邊最接近位置 >p 的
    後綴的 lcp · 0-based
117 // pre[i] = s[i] 離他左邊最接近位置 >p 的後綴的 lcp · 0-
    based
118 // suf[i] = s[i] 離他右邊最接近位置 >p 的後綴的 lcp · 0-
    based
119 // da12fa
120 pair<vector<int>, vector<int>> get_left_and_right_lcp(int
    p){
121     vector<int> pre(p+1);
122     vector<int> suf(p+1);
123
124     { // build pre
125         int now = 0;
126         for (int i=0 ; i<s.size() ; i++){
127             if (sa[i]<=p){
128                 pre[sa[i]] = now;
129                 if (i<lcp.size()) now = min(now, lcp[i]);
130             }else{
131                 if (i<lcp.size()) now = lcp[i];
132             }
133         }
134     }
135     { // build suf
136         int now = 0;
137         for (int i=s.size()-1 ; i>=0 ; i--){
138             if (sa[i]<=p){
139                 suf[sa[i]] = now;
140                 if (i-1>=0) now = min(now, lcp[i-1]);
141             }else{
142                 if (i-1>=0) now = lcp[i-1];
143             }
144         }
145     }
146
147     return {pre, suf};
148 }
149 };

```

8.8 Suffix Automata [efe54a]

```

1 // root node is 0
2 // node -> strings with the same endpos set,
3 //         length in range [len[fa] + 1, len]
4 // Link -> Longest suffix with different endpos set
5 // len -> longest path length from root
6 // cnt -> size of endpos set
7 // node's endpos set -> pos (1-based) in the subtree (Link)
  of node (Some node have pos = 0 and should be ignored)
8 struct SAM {
9     static constexpr int N = (int)(1e5 + 10), Z = 26;
10    int ch[2 * N][Z], len[2 * N], link[2 * N], pos[2 * N], cnt
        [2 * N], sz;
11    int newnode() {
12        fill_n(ch[sz], Z, 0);
13        len[sz] = link[sz] = pos[sz] = cnt[sz] = 0;
14        return sz++;
15    }
16    void build(string s) {
17        sz = 0, newnode(), link[0] = -1;
18        int lst = 0;
19        for (int i = 0; i < s.size(); ++i) {
20            int c = s[i] - 'a'; // you may need to change to 'A'
21            int cur = newnode();
22            len[cur] = len[lst] + 1, pos[cur] = i + 1;

```

```

23    int p = lst;
24    while (~p && !ch[p][c])
25        ch[p][c] = cur, p = link[p];
26    if (p == -1) link[cur] = 0;
27    else {
28        int q = ch[p][c];
29        if (len[p] + 1 == len[q]) {
30            link[cur] = q;
31        } else {
32            int nxt = newnode();
33            len[nxt] = len[p] + 1, link[nxt] = link[q];
34            pos[nxt] = 0;
35            for (int j = 0; j < Z; ++j)
36                ch[nxt][j] = ch[q][j];
37            while (~p && ch[p][c] == q)
38                ch[p][c] = nxt, p = link[p];
39            link[q] = link[cur] = nxt;
40        }
41    }
42    cnt[cur]++, lst = cur;
43 }
44 vector<int> p(sz);
45 iota(p.begin(), p.end(), 0);
46 sort(p.begin(), p.end(), [&](int i, int j) {
47     return len[i] > len[j];
48 });
49 for (int i = 0; i < sz; ++i) if (~link[p[i]])
50     cnt[link[p[i]]] += cnt[p[i]];
51 }
52 };

```

8.9 Wildcard Matching [e93074]

```

1 const int B = 27;
2 string p;
3 for (int i=0 ; i<B ; i++) p += ('a'+i);
4 p += '*';
5 vector<vector<double>> res(B);
6 for (int i=0 ; i<B ; i++){
7     vector<double> ss, tt;
8     for (auto x : s) ss.push_back(x==p[i] || x=='*');
9     for (auto x : t) tt.push_back(x==p[i] || x=='*');
10    reverse(tt.begin(), tt.end());
11    res[i] = PolyMul(ss, tt);
12 }
13 for (int i=t.size()-1 ; i+t.size()-1<res[0].size() ; i++){
14     int total = 0;
15     for (int j=0 ; j<B-1 ; j++) total += (int)abs(round(res[j
        ][i]));
16     total -= 25*(int)abs(round(res[26][i]));
17     cout << (total==t.size());
18 }

```

8.10 Z Algorithm [9d559a]

```

1 // z[i] 回傳 s[0...] 跟 s[i...] 的 lcp, z[0] = 0
2 vector<int> z_function(string s){
3     vector<int> z(s.size());
4     int l = -1, r = -1;
5     for (int i=1 ; i<s.size() ; i++){
6         z[i] = i>r ? 0 : min(r-i, z[i-1]);
7         while (i+z[i]<s.size() && s[i+z[i]]==s[z[i]]) z[i]++;
8         if (i+z[i]>r) l=i, r=i+z[i];
9     }
10    return z;

```